

MS SQL Server – Querying Movies Database using SQL

“Querying Movies Database using SQL” is a major assignment I completed following a bunch of smaller SQL assignments for my Business Intelligence course in the Big Data Analytics program. In this project, I use SQL in MS SQL Server to create databases, schemas, insert data with different attributes, create tables with different constraints and query the inserted data per tasks assigned by my course professor to ensure my competency using MS SQL Server.

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Perform the following tasks in SQL

1. Create a database “Movies”.
2. Create a “mov” schema under “Movies”.

Solution:

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure, including the 'Movies' database and the 'mov' schema. The Query Editor in the center contains the following SQL script:

```
--Final Project 2 - SQL: Group11

--1. Create a database "Movies".
Create database Movies;

use Movies

--2. Create a "mov" schema under "Movies".
Create SCHEMA mov;

USE Movies;

--3. Create a table call "Movie_Director "under mov schema with the following specifications-
--a. Movie_Director must have the following attribute.
--b. Add the following constraint.
-- i. Director_ID: Auto Increment, Primary Key, Not null, clustered index.
-- ii. CreatedOn: Not Null, Default as Server date.
--c. Insert the following records based on the following specifications.
-- i. Director_ID: Director ID must be starts from 100 and incremented by 10.
-- ii. CreatedOn: Created as default date as system date.

CREATE TABLE mov.Movie_Director (
    Director_ID INT IDENTITY(100,10) NOT NULL CONSTRAINT PK_Movie_Director PRIMARY KEY CLUSTERED,
    Director_First_Name VARCHAR(50),
    Director_Last_Name VARCHAR(50),
    Director_Age_in_Years INT,
    Director_Gender VARCHAR(10),
    CreatedOn DATE NOT NULL CONSTRAINT DF_CreatedOn DEFAULT GETDATE()
);
```

The Results pane at the bottom shows the data inserted into the 'Movie_Director' table:

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn	
8	170	Mark	Palansky	53	Male	2024-03-23
9	180	Jeff	Lowell	49	Male	2024-03-23
10	190	Simon	Curtis	37	Male	2024-03-23
11	200	Marc	Lawrence	95	Male	2024-03-23
12	210	Anand	Tucker	59	Male	2024-03-23
13	220	Judd	Apatow	55	Male	2024-03-23
14	230	Cary	Fukunaga	45	Male	2024-03-23
15	240	Mark	Helfrich	49	Male	2024-03-23
16	250	Nanette	Burstein	52	Female	2024-03-23
17	260	James	McAvoy	44	Male	2024-03-23
18	270	Mark	Waters	58	Male	2024-03-23
19	280	Seth	Gordon	46	Male	2024-03-23
20	290	Alex	Kendrick	52	Male	2024-03-23
21	300	Kevin	Lima	60	Male	2024-03-23
22	310	Lasse	Hallström	76	Male	2024-03-23
23	320	Ewan	McGregor	52	Male	2024-03-23
24	330	Rajkumar	Hirani	60	Male	2024-03-23
25	340	Ashutosh	Gowariker	59	Male	2024-03-23
26	350	Karan	Johar	50	Male	2024-03-23
27	360	S.S	Rajamouli	49	Male	2024-03-23
28	370	Sukumar	NULL	53	Male	2024-03-23
29	380	Aditya	Chopra	51	Male	2024-03-23
30	390	Umesh	Shukla	52	Male	2024-03-23

3. Create a table call “Movie_Director “under mov schema with the following specifications-
 - a. Movie_Director must have the following attribute.

Column Name	Data Type	Description
Director_ID	Integer	Director ID
Director_First_Name	Varchar	Director First Name
Director_Last_Name	Varchar	Director Last Name

Director_Age_in_Years	Integer	Director Age in Year
Director_Gender	Varchar	Date of Joining
CreatedOn	Date	Record Creation Date

b. Add the following constraint.

- i. Director_ID: Auto Increment, Primary Key, Not null, clustered index.
- ii. CreatedOn: Not Null, Default as Server date.

c. Insert the following records based on the following specifications.

- i. Director_ID: Director ID must be starts from 100 and incremented by 10.
- ii. CreatedOn: Created as default date as system date.

Solution:

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the 'Movies' database selected. The right pane shows the SQL Server Enterprise Query window with the following SQL script:

```
--Final Project 2 - SQL: Group11
--1. Create a database "Movies".
Create database Movies;

use Movies

--2. Create a "mov" schema under "Movies".
Create SCHEMA mov;

USE Movies;

--3. Create a table call "Movie_Director" under mov schema with the following specifications-
--a. Movie_Director must have the following attribute.
--b. Add the following constraint.
-- i. Director_ID: Auto Increment, Primary Key, Not null, clustered index.
-- ii. CreatedOn: Not Null, Default as Server date.
--c. Insert the following records based on the following specifications.
-- i. Director_ID: Director ID must be starts from 100 and incremented by 10.
-- ii. CreatedOn: Created as default date as system date.

CREATE TABLE mov.Movie_Director (
    Director_ID INT IDENTITY(100,10) NOT NULL CONSTRAINT PK_Movie_Director PRIMARY KEY CLUSTERED,
    Director_First_Name VARCHAR(50),
    Director_Last_Name VARCHAR(50),
    Director_Age_in_Years INT,
    Director_Gender VARCHAR(10),
    CreatedOn DATE NOT NULL CONSTRAINT DF_CreatedOn DEFAULT GETDATE()
);
```

The bottom pane shows the results of the query, displaying 30 rows of data inserted into the 'mov.Movie_Director' table. The columns are Director_ID, Director_First_Name, Director_Last_Name, Director_Age_in_Years, Director_Gender, and CreatedOn. The data is as follows:

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn	
8	170	Mark	Palansky	53	Male	2024-03-23
9	180	Jeff	Lowell	49	Male	2024-03-23
10	190	Simon	Curtis	37	Male	2024-03-23
11	200	Marc	Lawrence	95	Male	2024-03-23
12	210	Anand	Tucker	59	Male	2024-03-23
13	220	Judd	Apatow	55	Male	2024-03-23
14	230	Cary	Fukunaga	45	Male	2024-03-23
15	240	Mark	Helfrich	49	Male	2024-03-23
16	250	Nanette	Burnstein	52	Female	2024-03-23
17	260	James	McAvoy	44	Male	2024-03-23
18	270	Mark	Waters	58	Male	2024-03-23
19	280	Seth	Gordon	46	Male	2024-03-23
20	290	Alex	Kendrick	52	Male	2024-03-23
21	300	Kevin	Lima	60	Male	2024-03-23
22	310	Lasse	Hallström	76	Male	2024-03-23
23	320	Ewan	McGregor	52	Male	2024-03-23
24	330	Rajkumar	Hirani	60	Male	2024-03-23
25	340	Ashutosh	Gowariker	59	Male	2024-03-23
26	350	Karan	Johar	50	Male	2024-03-23
27	360	S.S	Rajamouli	49	Male	2024-03-23
28	370	Sukumar	NULL	53	Male	2024-03-23
29	380	Aditya	Chopra	51	Male	2024-03-23
30	390	Umesh	Shukla	52	Male	2024-03-23

d. After Insertion, Data looks like-

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender
100	Kevin	Smith	52	Male
110	Miguel	Arteta	41	Male
120	Mark	Johnson	58	Male
130	Tom	Vaughan	37	Male
140	Francis	Lawrence	52	Male
150	Adrienne	Shelly	40	Female
160	David	Slade	53	Male
170	Mark	Palansky	53	Male
180	Jeff	Lowell	49	Male
190	Simon	Curtis	37	Male
200	Marc	Lawrence	95	Male
210	Anand	Tucker	59	Male
220	Judd	Apatow	55	Male
230	Cary	Fukunaga	45	Male
240	Mark	Helfrich	49	Male
250	Nanette	Burstein	52	Female
260	James	McAvoy	44	Male
270	Mark	Waters	58	Male
280	Seth	Gordon	46	Male
290	Alex	Kendrick	52	Male
300	Kevin	Lima	60	Male
310	Lasse	Hallström	76	Male
320	Ewan	McGregor	52	Male
330	Rajkumar	Hirani	60	Male
340	Ashutosh	Gowariker	59	Male
350	Karan	Johar	50	Male
360	S.S	Rajamouli	49	Male
370	Sukumar	NULL	53	Male
380	Aditya	Chopra	51	Male
390	Umesh	Shukla	52	Male

Solution:

Object Explorer: XPS\SQLSERVER2022 (SQL Server 16.0.1110.1 - XPS\am...)

Query: `INSERT INTO mov.Movie_Director (Director_First_Name, Director_Last_Name, Director_Age_in_Years, Director_Gender) VALUES ('Kevin', 'Smith', 52, 'Male'), ('Miguel', 'Arteta', 41, 'Male'), ('Mark', 'Johnson', 58, 'Male'), ('Tom', 'Vaughan', 37, 'Male'), ('Francis', 'Lawrence', 52, 'Male'), ('Adrienne', 'Shelly', 40, 'Female'), ('David', 'Slade', 53, 'Male'), ('Mark', 'Palansky', 53, 'Male'), ('Jeff', 'Lowell', 49, 'Male'), ('Simon', 'Curtis', 37, 'Male'), ('Marc', 'Lawrence', 95, 'Male'), ('Anand', 'Tucker', 59, 'Male'), ('Judd', 'Apatow', 55, 'Male'), ('Cary', 'Fukunaga', 45, 'Male'), ('Mark', 'Helfrich', 49, 'Male'), ('Nanette', 'Burstein', 52, 'Female'), ('James', 'McAvoy', 44, 'Male'), ('Mark', 'Waters', 58, 'Male'), ('Seth', 'Gordon', 46, 'Male'), ('Alex', 'Kendrick', 52, 'Male'), ('Kevin', 'Lima', 60, 'Male'), ('Lasse', 'Hallström', 76, 'Male'), ('Ewan', 'McGregor', 52, 'Male'), ('Rajkumar', 'Hirani', 60, 'Male'), ('Ashutosh', 'Gowariker', 59, 'Male'), ('Karan', 'Johar', 50, 'Male'), ('S.S.', 'Rajamouli', 49, 'Male'), ('Sukumar', 'NULL', 53, 'Male'), ('Aditya', 'Chopra', 51, 'Male'), ('Umesh', 'Shukla', 52, 'Male');`

Select * from [mov].[Movie_Director];

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn	
8	170	Mark	Palansky	53	Male	2024-03-23
9	180	Jeff	Lowell	49	Male	2024-03-23
10	190	Simon	Curtis	37	Male	2024-03-23
11	200	Marc	Lawrence	95	Male	2024-03-23
12	210	Anand	Tucker	59	Male	2024-03-23
13	220	Judd	Apatow	55	Male	2024-03-23
14	230	Cary	Fukunaga	45	Male	2024-03-23
15	240	Mark	Helfrich	49	Male	2024-03-23
16	250	Nanette	Burstein	52	Female	2024-03-23
17	260	James	McAvoy	44	Male	2024-03-23
18	270	Mark	Waters	58	Male	2024-03-23
19	280	Seth	Gordon	46	Male	2024-03-23
20	290	Alex	Kendrick	52	Male	2024-03-23
21	300	Kevin	Lima	60	Male	2024-03-23
22	310	Lasse	Hallström	76	Male	2024-03-23
23	320	Ewan	McGregor	52	Male	2024-03-23
24	330	Rajkumar	Hirani	60	Male	2024-03-23

Query executed successfully.

2. Create a Movies table under mov schema with the following specifications-

a. Movies table must have the following attributes.

Column Name	Data Type	Description
Movie_ID	Integer	Director ID
Movie_Name	Varchar	Movie Name
Movie_Released_Year	Integer	Movie Released Year
Movie_Lead_Studio	Varchar	Movie Released By
Movie_Language	Varchar	Movie Language
Movie_Category	Varchar	Movie Category
Movie_Duration_in_Min	Integer	Movie Duration in minutes
Movie_Worldwide_Earning_in_-\$M	Decimal (15,2)	Movie Worldwide earnings
Movie_Type	Varchar	Movie Type
Director_ID	Integer	Director ID
CreatedOn	Date	Record Creation Date

b. Add the following constraint.

- i. Movie_ID: Auto Increment, Primary Key, Not null, clustered index.
- ii.CreatedOn: Not Null, Default as Server date.

c. Insert the following records based on the following specifications.

- i. Movie_ID: Movie ID must be starts from 1000 and incremented by 1.
- ii. CreatedOn: Created as default date as system date.
- iii. Director_ID: Foreign key from Movie_Director table.
- iv. Movie_Type: Movie Type as Hollywood and Bollywood.

Solution:

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0.1110.1 - XPS\ammmym (55))'. The 'mov' database is expanded, showing the 'Movie_Director' table and the 'Movies' table being created. The Query window in the center contains the following SQL script:

```
Select * from [mov].[Movie_Director];

--4. Create a Movies table under mov schema with the following specifications-
--a. Movies table must have the following attributes.
--b. Add the following constraint.
--i. Movie_ID: Auto Increment, Primary Key, Not null, clustered index.
--ii. CreatedOn: Not Null, Default as Server date.

--c. Insert the following records based on the following specifications.
--i. Movie_ID: Movie ID must be starts from 1000 and incremented by 1.
--ii. CreatedOn: Created as default date as system date.
--iii. Director_ID: Foreign key from Movie_Director table.
--iv. Movie_Type: Movie Type as Hollywood and Bollywood.

CREATE TABLE mov.Movies (
    Movie_ID INT IDENTITY(1000,1) NOT NULL CONSTRAINT PK_Movies PRIMARY KEY CLUSTERED,
    Movie_Name VARCHAR(50),
    Movie_Released_Year INT,
    Movie_Lead_Studio VARCHAR(50),
    Movie_Language VARCHAR(50),
    Movie_Category VARCHAR(50),
    Movie_Duration_in_Min INT,
    Movie_Worldwide_Earning_in_$M DECIMAL(15,2),
    Movie_Type VARCHAR(50),
    Director_ID INT CONSTRAINT FK_Movies_Director_ID FOREIGN KEY (Director_ID) REFERENCES mov.Movie_Director(Director_ID),
    CreatedOn DATE NOT NULL CONSTRAINT DF_Movies_CreatedOn DEFAULT GETDATE()
);
```

The Messages window at the bottom shows the successful execution of the query:

```
Commands completed successfully.

Completion time: 2024-03-23T19:51:40.2052727-04:00
```

The status bar at the bottom indicates that the query was executed successfully, with 0 rows returned.

Microsoft SQL Server Management Studio interface showing a query execution in the 'Movies Database'.

Object Explorer: Shows the database structure, including Tables, Views, and Security.

Query Window: Displays the following SQL query:

```
Director_ID INT CONSTRAINT FK_Movies_Director_ID FOREIGN KEY (Director_ID) REFERENCES mov.Movie_Director(Director_ID),
CreatedOn DATE NOT NULL CONSTRAINT DF_Movies_CreatedOn DEFAULT GETDATE()
);

--d. After Insertion, Table looks like as shown below-
INSERT INTO mov.Movies (Movie_Name, Movie_Released_Year, Movie_Lead_Studio, Movie_Language, Movie_Category, Movie_Duration_in_Min,
Movie_Worldwide_Earning_in_$M, Movie_Type, Director_ID)
VALUES
('Zack and Miri Make a Porno', 2008, 'The Weinstein Company', 'English', 'Romance', 101, 41.94, 'Hollywood', 100),
('Youth in Revolt', 2010, 'The Weinstein Company', 'English', 'Comedy', 90, 19.62, 'Hollywood', 110),
('When in Rome', 2010, 'Disney', 'English', 'Comedy', 91, 43.04, 'Hollywood', 120),
('What Happens in Vegas', 2008, 'Fox', 'English', 'Comedy', 99, 219.37, 'Hollywood', 130),
('Water for Elephants', 2011, '20th Century Fox', 'English', 'Drama', 120, 117.09, 'Hollywood', 140),
('Waitress', 2007, 'Independent', 'English', 'Romance', 108, 22.18, 'Hollywood', 150),
('Twilight', 2008, 'Summit', 'English', 'Romance', 122, 376.66, 'Hollywood', 160),
('Penelope', 2008, 'Summit', 'English', 'Comedy', 144, 20.74, 'Hollywood', 170),
('Over Her Dead Body', 2008, 'New Line', 'English', 'Comedy', 95, 20.71, 'Hollywood', 180),
('My Week with Marilyn', 2011, 'The Weinstein Company', 'English', 'Drama', 99, 8.26, 'Hollywood', 190),
('Music and Lyrics', 2007, 'Warner Bros.', 'English', 'Romance', 104, 145.9, 'Hollywood', 200),
('Leap Year', 2010, 'Universal', 'English', 'Comedy', 100, 32.59, 'Hollywood', 210),
('Knocked Up', 2007, 'Universal', 'English', 'Comedy', 129, 219, 'Hollywood', 220),
('Jane Eyre', 2011, 'Universal', 'English', 'Romance', 120, 30.15, 'Hollywood', 230),
('Good Luck Chuck', 2007, 'Lionsgate', 'English', 'Comedy', 101, 59.19, 'Hollywood', 240),
('Going the Distance', 2010, 'Warner Bros.', 'English', 'Comedy', 103, 42.05, 'Hollywood', 250),
('Gnomeo and Juliet', 2011, 'Disney', 'English', 'Animation', 84, 193.97, 'Hollywood', 260),
('Ghosts of Girlfriends Past', 2009, 'Warner Bros.', 'English', 'Comedy', 100, 102.22, 'Hollywood', 270),
('Four Christmases', 2008, 'Warner Bros.', 'English', 'Comedy', 88, 161.83, 'Hollywood', 280),
('Fireproof', 2008, 'Independent', 'English', 'Drama', 122, 33.47, 'Hollywood', 290),
('Enchanted', 2007, 'Disney', 'English', 'Comedy', 107, 340.49, 'Hollywood', 300),
('Dear John', 2010, 'Sony', 'English', 'Drama', 108, 114.97, 'Hollywood', 310),
('Beginners', 2011, 'Independent', 'English', 'Comedy', 105, 14.31, 'Hollywood', 320),
('3 Idiots', 2009, 'Vinod Chopra Films', 'Hindi', 'Comedy', 171, 4000, 'Bollywood', 330),
('Lagaan', 2001, 'Amir Khan Productions', 'Hindi', 'Romance', 224, 659, 'Bollywood', 340),
('My Name Is Khan', 2010, 'Dharma Productions', 'Hindi', 'Drama', 165, 48.77, 'Bollywood', 350),
('Baahubali', 2015, 'Arka Media Works', 'Telugu', 'Thriller', 159, 6500, 'Bollywood', 360),
('Kaho Rakhoo To Saavari' 2008, 'New Regency', 'Hindi', 'Romance', 100, 5000, 'Hollywood', 300);
```

Messages: (30 rows affected)
Completion time: 2024-03-23T20:02:41.9021988-04:00

Status Bar: Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 0 rows

d. After Insertion, Table looks like as shown below-

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID
1000	Zack and Miri Make a Porno	2008	The Weinstein Company	English	Romance	101	41.94	Hollywood	100
1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110
1002	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120
1003	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130
1004	Water For Elephants	2011	20th Century Fox	English	Drama	120	117.09	Hollywood	140
1005	Waitress	2007	Independent	English	Romance	108	22.18	Hollywood	150
1006	Twilight	2008	Summit	English	Romance	122	376.66	Hollywood	160
1007	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170
1008	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180
1009	My Week with Marilyn	2011	The Weinstein Company	English	Drama	99	8.26	Hollywood	190
1010	Music and Lyrics	2007	Warner Bros.	English	Romance	104	145.9	Hollywood	200
1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210
1012	Knocked Up	2007	Universal	English	Comedy	129	219	Hollywood	220
1013	Jane Eyre	2011	Universal	English	Romance	120	30.15	Hollywood	230
1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240
1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250
1016	Gnomeo and Juliet	2011	Disney	English	Animation	84	193.97	Hollywood	260
1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270
1018	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280
1019	Fireproof	2008	Independent	English	Drama	122	33.47	Hollywood	290
1020	Enchanted	2007	Disney	English	Comedy	107	340.49	Hollywood	300
1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310
1022	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320
1023	3 Idiots	2009	Vinod Chopra Films	Hindi	Comedy	171	4000	Bollywood	330
1024	Lagaan	2001	Aamir Khan Productions	Hindi	Romance	224	659	Bollywood	340
1025	My Name Is Khan	2010	Dharma Productions	Hindi	Drama	165	48.77	Bollywood	350
1026	Baahubali	2015	Arka Media Works	Telugu	Thriller	159	6500	Bollywood	360
1027	Dilwale Dulhania Le Jayenge	1995	Yash Chopra	Hindi	Romance	189	2000	Bollywood	380
1028	Oh My God	2012		Hindi	Comedy	165	1200	Bollywood	390
1029	Pushpa	2021	Mythri Movie Makers	Telugu	Drama	179	3730	Bollywood	370

Solution:

The screenshot shows the Microsoft SQL Server Management Studio interface. The Query Editor contains a SQL query that selects various details for 30 movies from the [mov].[Movies] table. The Results pane displays the output of this query, showing columns for Movie_ID, Movie_Name, Movie_Released_Year, Movie_Lead_Studio, Movie_Language, Movie_Category, Movie_Duration_in_Min, Movie_Worldwide_Earning_in_\$M, Movie_Type, Director_ID, and CreatedOn. The status bar at the bottom indicates that the query was executed successfully.

```

('Gnomeo and Juliet', 2011, 'Disney', 'English', 'Animation', 84, 193.97, 'Hollywood', 260),
('Ghosts of Girlfriends Past', 2009, 'Warner Bros.', 'English', 'Comedy', 100, 102.22, 'Hollywood', 270),
('Four Christmases', 2008, 'Warner Bros.', 'English', 'Comedy', 88, 161.83, 'Hollywood', 280),
('Fireproof', 2008, 'Independent', 'English', 'Drama', 122, 33.47, 'Hollywood', 290),
('Enchanted', 2007, 'Disney', 'English', 'Comedy', 107, 340.49, 'Hollywood', 300),
('Dear John', 2010, 'Sony', 'English', 'Drama', 108, 114.97, 'Hollywood', 310),
('Beginners', 2011, 'Independent', 'English', 'Comedy', 105, 14.31, 'Hollywood', 320),
('3 Idiots', 2009, 'Vinod Chopra Films', 'Hindi', 'Comedy', 171, 4000, 'Bollywood', 330),
('Lagaan', 2001, 'Aamir Khan Productions', 'Hindi', 'Romance', 224, 659, 'Bollywood', 340),
('My Name Is Khan', 2010, 'Dharma Productions', 'Hindi', 'Drama', 165, 48.77, 'Bollywood', 350),
('Baahubali', 2015, 'Arka Media Works', 'Telugu', 'Thriller', 159, 6500, 'Bollywood', 360),
('Dilwale Dulhania Le Jayenge', 1995, 'Yash Chopra', 'Hindi', 'Romance', 189, 2000, 'Bollywood', 380),
('Oh My God', 2012, '', 'Hindi', 'Comedy', 165, 1200, 'Bollywood', 390),
('Pushpa', 2021, 'Mythri Movie Makers', 'Telugu', 'Drama', 179, 3730, 'Bollywood', 370);

Select * from [mov].[Movies];
  
```

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	CreatedOn	
1	1000	Zack and Miri Make a Porno	2008	The Weinstein Company	English	Romance	101	41.94	Hollywood	100	2024-03-23
2	1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110	2024-03-23
3	1002	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120	2024-03-23
4	1003	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130	2024-03-23
5	1004	Water For Elephants	2011	20th Century Fox	English	Drama	120	117.09	Hollywood	140	2024-03-23
6	1005	Waitress	2007	Independent	English	Romance	108	22.18	Hollywood	150	2024-03-23
7	1006	Twilight	2008	Summit	English	Romance	122	376.66	Hollywood	160	2024-03-23
8	1007	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170	2024-03-23
9	1008	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180	2024-03-23
10	1009	My Week with Marilyn	2011	The Weinstein Company	English	Drama	99	8.26	Hollywood	190	2024-03-23
11	1010	Music and Lyrics	2007	Warner Bros.	English	Romance	104	145.90	Hollywood	200	2024-03-23
12	1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210	2024-03-23
13	1012	Knocked Up	2007	Universal	English	Comedy	129	219.00	Hollywood	220	2024-03-23
14	1013	Jane Eyre	2011	Universal	English	Romance	120	30.15	Hollywood	230	2024-03-23
15	1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240	2024-03-23
16	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250	2024-03-23
17	1016	Gnomeo and Juliet	2011	Disney	English	Animation	84	193.97	Hollywood	260	2024-03-23
18	1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270	2024-03-23
19	1018	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280	2024-03-23
20	1019	Fireproof	2008	Independent	English	Drama	122	33.47	Hollywood	290	2024-03-23
21	1020	Enchanted	2007	Disney	English	Comedy	107	340.49	Hollywood	300	2024-03-23
22	1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310	2024-03-23
23	1022	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320	2024-03-23
24	1023	3 Idiots	2009	Vinod Chopra Films	Hindi	Comedy	171	4000.00	Bollywood	330	2024-03-23
25	1024	Lagaan	2001	Aamir Khan Productions	Hindi	Romance	224	659.00	Bollywood	340	2024-03-23
26	1025	My Name Is Khan	2010	Dharma Productions	Hindi	Drama	165	48.77	Bollywood	350	2024-03-23
27	1026	Baahubali	2015	Arka Media Works	Telugu	Thriller	159	6500.00	Bollywood	360	2024-03-23
28	1027	Dilwale Dulhania Le Jayen...	1995	Yash Chopra	Hindi	Romance	189	2000.00	Bollywood	380	2024-03-23
29	1028	Oh My God	2012		Hindi	Comedy	165	1200.00	Bollywood	390	2024-03-23
30	1029	Pushpa	2021	Mythri Movie Makers	Telugu	Drama	179	3730.00	Bollywood	370	2024-03-23

3. Create a Movie_Actor table under mov schema with the following specifications-

a. Movie_Actor table must have the following attributes.

Column Name	Data Type	Description
Actor_ID	Integer	Actor ID
Actor_First_Name	Varchar	Actor First Name
Actor_Last_Name	Varchar	Actor Last Name
Actor_Age_in_Years	Integer	Actor Age in Years
Actor_Location	Varchar	Actor Location
Movie_ID	Integer	Movie ID
CreatedOn	Date	Record Creation Date

- b. Add the following constraint.
- Actor_ID: Auto Increment, Primary Key, Not null, clustered index.
 - CreatedOn: Not Null, Default as Server date.
- c. Insert the following records based on the following specifications.
- Actor_ID: Actor ID must be starts from 10 and incremented by 1.
 - CreatedOn: Created as default date as system date.
 - Movie_ID: Foreign key from Movies table.

Solution:

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The right pane shows the 'Query Editor' with the following SQL script:

```
USE [Movies]
GO

--c. Insert the following records based on the following specifications.
--i. Actor_ID: Actor ID must be starts from 10 and incremented by 1.
--ii. CreatedOn: Created as default date as system date.
--iii. Movie_ID: Foreign key from Movies table.

--5. Create a Movie_Actor table under mov schema with the following specifications-
--a. Movie_Actor table must have the following attributes.
--b. Add the following constraint.
--i. Actor_ID: Auto Increment, Primary Key, Not null, clustered index.
--ii. CreatedOn: Not Null, Default as Server date.

--c. Insert the following records based on the following specifications.
--i. Actor_ID: Actor ID must be starts from 10 and incremented by 1.
--ii. CreatedOn: Created as default date as system date.
--iii. Movie_ID: Foreign key from Movies table.

CREATE TABLE mov.Movie_Actor (
    Actor_ID INT IDENTITY(10,1) NOT NULL CONSTRAINT PK_Movie_Actor PRIMARY KEY CLUSTERED,
    Actor_First_Name VARCHAR(50),
    Actor_Last_Name VARCHAR(50),
    Actor_Age_in_Years INT,
    Actor_Location VARCHAR(50),
    Movie_ID INT CONSTRAINT FK_Movie_ID FOREIGN KEY (Movie_ID) REFERENCES mov.Movies(Movie_ID),
    CreatedOn DATE NOT NULL CONSTRAINT DF_Movie_Actor_CreatedOn DEFAULT GETDATE()
);

--d. After Insertion, Table looks like as shown below.
```

The bottom pane shows the 'Messages' window with the following output:

```
Commands completed successfully.
Completion time: 2024-03-23T20:22:26.4983470-04:00
```

The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

Microsoft SQL Server Management Studio interface showing a query execution in the 'Movies' database.

Object Explorer: Shows the database structure for 'XPS\SQLSERVER2022 (SQL S...)' including Databases, System Databases, Database Snapshots, Movies, Database Diagrams, Tables, System Table, FileTables, External Table, Graph Tables, mov.Movie_I, mov.Movies, Dropped Led, Views, External Resource, Synonyms, Programmability, Query Store, Service Broker, Storage, Security, Server Objects, Replication, Always On High Avail, Management, Integration Services Cat, SQL Server Agent (Ager), and XEvent Profiler.

Query Editor: Displays the following SQL query:

```
--d. After Insertion, Table looks like as shown below-
INSERT INTO mov.Movie_Actor (Actor_First_Name, Actor_Last_Name, Actor_Age_in_Years, Actor_Location, Movie_ID)
VALUES
('Seth', 'Rogen', 53, 'Los Angeles', 1000),
('Michael', 'Cera', 49, 'New York', 1001),
('Josh', 'Duhamel', 37, 'North Dakota', 1002),
('Jason', 'Sudeikis', 60, 'Kansas', 1003),
('Robert', 'Pattinson', 45, 'Los Angeles', 1004),
('Nathan', 'Phillip', 55, 'Canada', 1005),
('Robert', 'Pattinson', 45, 'Los Angeles', 1006),
('James', 'McAvoy', 49, 'Scotland', 1007),
('Paul', 'Rudd', 52, 'New York', 1008),
('Kenneth', 'Branagh', 44, 'Northern Ireland', 1009),
('Hugh', 'Grant', 58, 'London', 1010),
('Matthew', 'Goode', 46, 'England', 1011),
('Judd', 'Apatow', 58, 'Los Angeles', 1012),
('Michael', 'Fassbender', 46, 'Germany', 1013),
('Dane', 'Cook', 52, 'United States', 1014),
('Jason', 'Sudeikis', 60, 'Kansas', 1015),
('Kelly', 'Asbury', 76, 'United States', 1016),
('Matthew', 'McConaughey', 52, 'Los Angeles', 1017),
('Vince', 'Vaughn', 60, 'Minnesota', 1018),
('Kirk', 'Cameron', 59, 'United States', 1019),
('James', 'Marsden', 50, 'Columbia', 1020),
('Channing', 'Tatum', 58, 'Alabama', 1021),
('Mike', 'Mills', 37, 'New York', 1022),
('Aamir', 'Khan', 52, 'India', 1023),
('Aamir', 'Khan', 52, 'India', 1024),
('Shah Rukh', 'Khan', 53, 'India', 1025),
('Prabhas', 'NULL', 53, 'India', 1026),
('Allu', 'Arjun', 49, 'India', 1027),
('Shah Rukh', 'Khan', 53, 'India', 1028),
('Akshay', 'Kumar', 50, 'India', 1029);
```

Messages: (30 rows affected)
Completion time: 2024-03-23T20:26:08.7395200-04:00

Status Bar: Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 0 rows

d. After Insertion, Table looks like as shown below-

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID
10	Seth	Rogen	53	Los Angeles	1000
11	Michael	Cera	49	New York	1001
12	Josh	Duhamel	37	North Dakota	1002
13	Jason	Sudeikis	60	Kansas	1003
14	Robert	Pattinson	45	Los Angeles	1004
15	Nathan	Fillion	55	Canada	1005
16	Robert	Pattinson	45	Los Angeles	1006
17	James	McAvoy	49	Scotland	1007
18	Paul	Rudd	52	New York	1008
19	Kenneth	Branagh	44	Northern Ireland	1009
20	Hugh	Grant	58	London	1010
21	Matthew	Goode	46	England	1011
22	Judd	Apatow	58	Los Angeles	1012
23	Michael	Fassbender	46	Germany	1013
24	Dane	Cook	52	United States	1014
25	Jason	Sudeikis	60	Kansas	1015
26	Kelly	Asbury	76	United States	1016
27	Matthew	McConaughey	52	Los Angeles	1017
28	Vince	Vaughn	60	Minnesota	1018
29	Kirk	Cameron	59	United States	1019
30	James	Marsden	50	Columbia	1020
31	Channing	Tatum	58	Alabama	1021
32	Mike	Mills	37	New York	1022
33	Aamir	Khan	52	India	1023
34	Aamir	Khan	52	India	1024
35	Shah Rukh	Khan	53	India	1025
36	Prabhas	NULL	53	India	1026
37	Allu	Arjun	49	India	1027
38	Shah Rukh	Khan	53	India	1028
39	Akshay	Kumar	50	India	1029

Solution:

Query executed successfully.

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn
10	Seth	Rogen	53	Los Angeles	1000	2024-03-23
11	Michael	Cera	49	New York	1001	2024-03-23
12	Josh	Duhamel	37	North Dakota	1002	2024-03-23
13	Jason	Sudekhis	60	Kansas	1003	2024-03-23
14	Robert	Pattinson	45	Los Angeles	1004	2024-03-23
15	Nathan	Fillon	55	Canada	1005	2024-03-23
16	Robert	Pattinson	45	Los Angeles	1006	2024-03-23
17	James	McAvoy	49	Scotland	1007	2024-03-23
18	Paul	Rudd	52	New York	1008	2024-03-23
19	Kenneth	Branagh	44	Northern Ireland	1009	2024-03-23
20	Hugh	Grant	58	London	1010	2024-03-23
21	Matthew	Goode	46	England	1011	2024-03-23
22	Judd	Apatow	58	Los Angeles	1012	2024-03-23
23	Michael	Fassbender	46	Germany	1013	2024-03-23
24	Dane	Cook	52	United States	1014	2024-03-23
25	Jason	Sudekhis	60	Kansas	1015	2024-03-23
26	Kelly	Asbury	76	United States	1016	2024-03-23
27	Matthew	McConaughey	52	Los Angeles	1017	2024-03-23
28	Vince	Vaughn	60	Minnesota	1018	2024-03-23
29	Kirk	Cameron	59	United States	1019	2024-03-23
30	James	Marsden	50	Columbia	1020	2024-03-23
31	Channing	Tatum	58	Alabama	1021	2024-03-23
32	Mike	Mills	37	New York	1022	2024-03-23
33	Aamir	Khan	52	India	1023	2024-03-23
34	Aamir	Khan	52	India	1024	2024-03-23
35	Shah Rukh	Khan	53	India	1025	2024-03-23
36	Prabhas	NULL	53	India	1026	2024-03-23
37	Allu	Arjun	49	India	1027	2024-03-23
38	Shah Rukh	Khan	53	India	1028	2024-03-23
39	Akshay	Kumar	50	India	1029	2024-03-23

4. Create a Movie_Rating table under mov schema with the following specifications-

a. Movie_Rating table must have the following attributes.

Column Name	Data Type	Description
Movie_Rating_ID	Integer	Movie Rating ID
Rating_Audience_Score	Varchar	Actor First Name
Rating_Rotten_Tomatoes	Varchar	Actor Last Name
Movie_ID	Integer	Movie ID
CreatedOn	Date	Record Creation Date

b. Add the following constraint.

- i. Movie_Rating_ID: Auto Increment, Primary Key, Not null, clustered index.
- ii. CreatedOn: Not Null, Default as Server date.

c. Insert the following records based on the following specifications.

- iv. Movie_Rating_ID: Actor ID must be starts from 1 and incremented by 1.
- v. CreatedOn: Created as default date as system date.
- vi. Movie_ID: Foreign key from Movies table.

Solution:

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL)' and 'Movies'. The main window shows a SQL script with the following content:

```
Select * from [mov].[Movie_Actor];

--6. Create a Movie_Rating table under mov schema with the following specifications-
--a. Movie_Rating table must have the following attributes.

--b. Add the following constraint.
--i. Movie_Rating_ID: Auto Increment, Primary Key, Not null, clustered index.
--ii. CreatedOn: Not Null, Default as Server date.

--c. Insert the following records based on the following specifications.
--iv. Movie_Rating_ID: Actor ID must be starts from 1 and incremented by 1.
--v. CreatedOn: Created as default date as system date.
--vi. Movie_ID: Foreign key from Movies table.

CREATE TABLE mov.Movie_Rating (
    Movie_Rating_ID INT IDENTITY(1,1) NOT NULL CONSTRAINT PK_Movie_Rating PRIMARY KEY CLUSTERED,
    Rating_Audience_Score VARCHAR(10),
    Rating_Rotten_Tomatoes VARCHAR(10),
    Movie_ID INT CONSTRAINT FK_Movie_Rating_Movie_ID FOREIGN KEY (Movie_ID) REFERENCES mov.Movies(Movie_ID),
    CreatedOn DATE NOT NULL CONSTRAINT DF_Movie_Rating_CreatedOn DEFAULT GETDATE()
);
```

The Messages pane at the bottom shows the following output:

```
Commands completed successfully.
Completion time: 2024-03-23T20:35:26.8907247-04:00
```

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 0 rows'.

Microsoft SQL Server Management Studio interface showing a query execution in the 'Movies Database'.

Object Explorer: Shows the database structure including Databases, Tables, Views, and Security.

Query Window: Displays the following SQL query:

```
--d. After Insertion, Table looks like as shown below-  
INSERT INTO mov.Movie_Rating (Rating_Audience_Score, Rating_Rotten_Tomatoes, Movie_ID)  
VALUES  
(70, 64, 1000),  
(52, 68, 1001),  
(44, 15, 1002),  
(72, 28, 1003),  
(72, 68, 1004),  
(67, 89, 1005),  
(82, 49, 1006),  
(74, 52, 1007),  
(47, 15, 1008),  
(84, 83, 1009),  
(70, 63, 1010),  
(49, 21, 1011),  
(83, 91, 1012),  
(77, 85, 1013),  
(61, 3, 1014),  
(56, 53, 1015),  
(52, 56, 1016),  
(47, 27, 1017),  
(52, 26, 1018),  
(51, 40, 1019),  
(80, 93, 1020),  
(66, 29, 1021),  
(80, 84, 1022),  
(95, 100, 1023),  
(81, 95, 1024),  
(79, 83, 1025),  
(80, 90, 1026),  
(76, 82, 1027),  
(85, 100, 1028),  
(81, 74, 1029);
```

Messages Window: Shows the execution result:

```
(30 rows affected)  
Completion time: 2024-03-23T20:40:28.1079712-04:00
```

Status Bar: Query executed successfully. | XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 0 rows

d. After Insertion, Table looks like as shown below-

Movie_Rating_ID	Rating_Audience_Score	Rating_Rotten_Tomatoes	Movie_ID
1	70	64	1000
2	52	68	1001
3	44	15	1002
4	72	28	1003
5	72	60	1004
6	67	89	1005
7	82	49	1006
8	74	52	1007
9	47	15	1008
10	84	83	1009
11	70	63	1010
12	49	21	1011
13	83	91	1012
14	77	85	1013
15	61	3	1014
16	56	53	1015
17	52	56	1016
18	47	27	1017
19	52	26	1018
20	51	40	1019
21	80	93	1020
22	66	29	1021
23	80	84	1022
24	95	100	1023
25	81	95	1024
26	79	83	1025
27	80	90	1026
28	76	82	1027
29	85	100	1028
30	81	74	1029

Solution:

Microsoft SQL Server Management Studio window titled "Movies Database - Final Project PT 2.sql - XPS\SQLSERVER2022\Movies (XPS\ammym (55))". The Object Explorer on the left shows the database structure. The query editor in the center contains the following SQL query:

```
('61', '3', 1014),  
('56', '53', 1015),  
('52', '56', 1016),  
('47', '27', 1017),  
('52', '26', 1018),  
('51', '40', 1019),  
('88', '93', 1020),  
('66', '29', 1021),  
('89', '84', 1022),  
('95', '100', 1023),  
('81', '95', 1024),  
('79', '83', 1025),  
('80', '90', 1026),  
('76', '82', 1027),  
('85', '100', 1028),  
('81', '74', 1029);  
  
Select * from [mov].[Movie_Rating];
```

The Results pane at the bottom displays the following data:

	Movie_Rating_ID	Rating_Audience_Score	Rating_Rotten_Tomatoes	Movie_ID	CreatedOn
1	1	70	64	1000	2024-03-23
2	2	52	68	1001	2024-03-23
3	3	44	15	1002	2024-03-23
4	4	72	28	1003	2024-03-23
5	5	72	60	1004	2024-03-23
6	6	67	89	1005	2024-03-23
7	7	82	49	1006	2024-03-23
8	8	74	52	1007	2024-03-23
9	9	47	15	1008	2024-03-23
10	10	84	83	1009	2024-03-23
11	11	70	63	1010	2024-03-23
12	12	49	21	1011	2024-03-23
13	13	83	91	1012	2024-03-23
14	14	77	85	1013	2024-03-23
15	15	61	3	1014	2024-03-23
16	16	56	53	1015	2024-03-23
17	17	52	56	1016	2024-03-23
18	18	47	27	1017	2024-03-23
19	19	52	26	1018	2024-03-23
20	20	51	40	1019	2024-03-23
21	21	80	93	1020	2024-03-23
22	22	66	29	1021	2024-03-23
23	23	80	84	1022	2024-03-23
24	24	95	100	1023	2024-03-23
25	25	81	95	1024	2024-03-23
26	26	79	83	1025	2024-03-23
27	27	80	90	1026	2024-03-23
28	28	76	82	1027	2024-03-23
29	29	85	100	1028	2024-03-23
30	30	81	74	1029	2024-03-23

The status bar at the bottom indicates "Query executed successfully." and "30 rows".

5. Write the following Query based on the above datasets.
- a. List all the Movies information from the Movies table.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a query window with the following SQL code:

```
SELECT * FROM [mov].[Movie_Rating];

--7. Write the following Query based on the above datasets.
--a. List all the Movies information from the Movies table.
SELECT * FROM [mov].[Movies];

--b. List all the Director information from the Director table.

--c. List all the Actor information from the Actor table.

--d. List all the Rating information from the Rating table.
```

The bottom pane shows the 'Results' tab with a table containing 29 rows of movie data. The status bar at the bottom indicates 'Query executed successfully.' and '30 rows'.

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	
1	1000	Zack and Mini Make a Pomo	2008	The Weinstein Company	English	Romance	101	41.94	Hollywood	100
2	1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110
3	1002	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120
4	1003	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130
5	1004	Water For Elephants	2011	20th Century Fox	English	Drama	120	117.09	Hollywood	140
6	1005	Watrees	2007	Independent	English	Romance	108	22.18	Hollywood	150
7	1006	Twilight	2008	Summit	English	Romance	122	376.66	Hollywood	160
8	1007	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170
9	1008	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180
10	1009	My Week with Marilyn	2011	The Weinstein Company	English	Drama	99	8.26	Hollywood	190
11	1010	Music and Lyrics	2007	Warner Bros.	English	Romance	104	145.90	Hollywood	200
12	1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210
13	1012	Knocked Up	2007	Universal	English	Comedy	129	219.00	Hollywood	220
14	1013	Jane Eyre	2011	Universal	English	Romance	120	30.15	Hollywood	230
15	1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240
16	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250
17	1016	Gnomeo and Juliet	2011	Disney	English	Animation	84	193.97	Hollywood	260
18	1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270
19	1018	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280
20	1019	Fireproof	2008	Independent	English	Drama	122	33.47	Hollywood	290
21	1020	Enchanted	2007	Disney	English	Comedy	107	340.49	Hollywood	300
22	1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310
23	1022	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320
24	1023	3 Idiots	2009	Vinod Chopra Films	Hindi	Comedy	171	4000.00	Bollywood	330
25	1024	Lagaan	2001	Aamir Khan Productions	Hindi	Romance	224	659.00	Bollywood	340
26	1025	My Name Is Khan	2010	Dharma Productions	Hindi	Drama	165	48.77	Bollywood	350
27	1026	Baahubali	2015	Arka Media Works	Telugu	Thriller	159	6500.00	Bollywood	360
28	1027	Dilwale Dulhania Le Jayenge	1995	Yash Chopra	Hindi	Romance	189	2000.00	Bollywood	380
29	1028	Oh My God	2012		Hindi	Comedy	165	1200.00	Bollywood	390

b. List all the Director information from the Director table.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The 'Query Editor' in the center contains a SQL script with comments and a query to list director information. The 'Results' pane at the bottom shows the output of the query, which is a table of director details.

```
--7. Write the following Query based on the above datasets.
--a. List all the Movies information from the Movies table.
SELECT * FROM [mov].[Movies];

--b. List all the Director information from the Director table.
SELECT * FROM [mov].[Movie_Director];

--c. List all the Actor information from the Actor table.

--d. List all the Rating information from the Rating table.
```

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn
1	Kevin	Smith	52	Male	2024-03-23
2	Miguel	Arteta	41	Male	2024-03-23
3	Mark	Johnson	58	Male	2024-03-23
4	Tom	Vaughan	37	Male	2024-03-23
5	Francis	Lawrence	52	Male	2024-03-23
6	Adrienne	Shelly	40	Female	2024-03-23
7	David	Slade	53	Male	2024-03-23
8	Mark	Palansky	53	Male	2024-03-23
9	Jeff	Lowell	49	Male	2024-03-23
10	Simon	Curtis	37	Male	2024-03-23
11	Marc	Lawrence	95	Male	2024-03-23
12	Anand	Tucker	59	Male	2024-03-23
13	Judd	Apatow	55	Male	2024-03-23
14	Cary	Fukunaga	45	Male	2024-03-23
15	Mark	Helfrich	49	Male	2024-03-23
16	Nanette	Burstein	52	Female	2024-03-23
17	James	McAvoy	44	Male	2024-03-23
18	Mark	Waters	58	Male	2024-03-23
19	Seth	Gordon	46	Male	2024-03-23
20	Alex	Kendrick	52	Male	2024-03-23
21	Kevin	Lima	60	Male	2024-03-23
22	Lasse	Hallström	76	Male	2024-03-23
23	Ewan	McGregor	52	Male	2024-03-23
24	Rajkumar	Hirani	60	Male	2024-03-23
25	Ashutosh	Gowariker	59	Male	2024-03-23
26	Karan	Johar	50	Male	2024-03-23
27	S.S	Rajamouli	49	Male	2024-03-23
28	Sukumar	NULL	53	Male	2024-03-23
29	Aditya	Chopra	51	Male	2024-03-23
30	Imesh	Shukla	52	Male	2024-03-23

c. List all the Actor information from the Actor table.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a SQL query window with the following code:

```
SELECT * FROM [mov].[Movie_Rating];

--7. Write the following Query based on the above datasets.
--a. List all the Movies information from the Movies table.
SELECT * FROM [mov].[Movies];

--b. List all the Director information from the Director table.
SELECT * FROM [mov].[Movie_Director];

--c. List all the Actor information from the Actor table.
SELECT * FROM mov.Movie_Actor;

--d. List all the Rating information from the Rating table.
```

The bottom pane shows the 'Results' tab with a table of actor information:

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn
1	Seth	Rogen	53	Los Angeles	1000	2024-03-23
2	Michael	Cera	49	New York	1001	2024-03-23
3	Josh	Duhamel	37	North Dakota	1002	2024-03-23
4	Jason	Sudekia	60	Kansas	1003	2024-03-23
5	Robert	Pattinson	45	Los Angeles	1004	2024-03-23
6	Nathan	Fillon	55	Canada	1005	2024-03-23
7	Robert	Pattinson	45	Los Angeles	1006	2024-03-23
8	James	McAvoy	49	Scotland	1007	2024-03-23
9	Paul	Rudd	52	New York	1008	2024-03-23
10	Kenneth	Branagh	44	Northern Ireland	1009	2024-03-23
11	Hugh	Grant	58	London	1010	2024-03-23
12	Matthew	Goode	46	England	1011	2024-03-23
13	Judd	Apatow	58	Los Angeles	1012	2024-03-23
14	Michael	Fassbender	46	Germany	1013	2024-03-23
15	Dane	Cook	52	United States	1014	2024-03-23
16	Jason	Sudekia	60	Kansas	1015	2024-03-23
17	Kelly	Asbury	76	United States	1016	2024-03-23
18	Matthew	McConaughey	52	Los Angeles	1017	2024-03-23
19	Vince	Vaughn	60	Minnesota	1018	2024-03-23
20	Kirk	Cameron	59	United States	1019	2024-03-23
21	James	Marsden	50	Columbia	1020	2024-03-23
22	Channing	Tatum	58	Alabama	1021	2024-03-23
23	Mike	Mills	37	New York	1022	2024-03-23
24	Aamir	Khan	52	India	1023	2024-03-23
25	Aamir	Khan	52	India	1024	2024-03-23
26	Shah Rukh	Khan	53	India	1025	2024-03-23
27	Prabhas	NULL	53	India	1026	2024-03-23
28	Allu	Ajun	49	India	1027	2024-03-23
29	Shah Rukh	Khan	53	India	1028	2024-03-23
30	Akshay	Kumar	50	India	1029	2024-03-23

The status bar at the bottom indicates 'Query executed successfully.' and '30 rows'.

d. List all the Rating information from the Rating table.

The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The title bar indicates the connection to 'Movies Database - Final Project PT 2.sql - XPS\SQLSERVER2022\Movies (XPS\ammym (55))'. The Object Explorer on the left shows the database structure, including tables like 'mov.Movie_Rating'. The central query editor contains the following SQL script:

```
SELECT * FROM [mov].[Movies];

--b. List all the Director information from the Director table.
SELECT * FROM [mov].[Movie_Director];

--c. List all the Actor information from the Actor table.
SELECT * FROM mov.Movie_Actor;

--d. List all the Rating information from the Rating table.
SELECT * FROM mov.Movie_Rating;

--e. List all the movie released in year "2010".

--f. List all the movie released by "The Weinstein Company" studio.

--g. List all the movie released in "English".
```

The 'Results' pane at the bottom shows the output of the query, displaying 30 rows of data from the 'Movie_Rating' table. The columns are 'Movie_Rating_ID', 'Rating_Audience_Score', 'Rating_Rotten_Tomatoes', 'Movie_ID', and 'CreatedOn'.

Movie_Rating_ID	Rating_Audience_Score	Rating_Rotten_Tomatoes	Movie_ID	CreatedOn
1	70	64	1000	2024-03-23
2	52	68	1001	2024-03-23
3	44	15	1002	2024-03-23
4	72	28	1003	2024-03-23
5	72	60	1004	2024-03-23
6	67	89	1005	2024-03-23
7	82	49	1006	2024-03-23
8	74	52	1007	2024-03-23
9	47	15	1008	2024-03-23
10	84	83	1009	2024-03-23
11	70	63	1010	2024-03-23
12	49	21	1011	2024-03-23
13	83	91	1012	2024-03-23
14	77	85	1013	2024-03-23
15	61	3	1014	2024-03-23
16	56	53	1015	2024-03-23
17	52	56	1016	2024-03-23
18	47	27	1017	2024-03-23
19	52	26	1018	2024-03-23
20	51	40	1019	2024-03-23
21	80	93	1020	2024-03-23
22	66	29	1021	2024-03-23
23	80	84	1022	2024-03-23
24	95	100	1023	2024-03-23
25	81	95	1024	2024-03-23
26	79	83	1025	2024-03-23
27	80	90	1026	2024-03-23
28	76	82	1027	2024-03-23
29	85	100	1028	2024-03-23
30	81	74	1029	2024-03-23

The status bar at the bottom indicates 'Query executed successfully.' and shows the current position in the query (Ln 255, Col 1, Ch 1, INS) and the number of rows returned (30 rows).

e. List all the movie released in year “2010”.

The screenshot displays the Microsoft SQL Server Management Studio interface. The left pane shows the Object Explorer with the 'Movies' database selected. The central query editor contains a SQL script with several queries, with the fifth query highlighted: `SELECT * FROM mov.Movies WHERE Movie_Released_Year = 2010;`. The bottom pane shows the results of this query as a table with 6 rows. A status bar at the bottom indicates 'Query executed successfully.' and '6 rows'.

	Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	CreatedOn
1	1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110	2024-03-23
2	1002	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120	2024-03-23
3	1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210	2024-03-23
4	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250	2024-03-23
5	1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310	2024-03-23
6	1025	My Name Is Khan	2010	Dharma Productions	Hindi	Drama	165	48.77	Bollywood	350	2024-03-23

f. List all the movie released by “The Weinstein Company” studio.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure, including the 'Movies' database and its tables. The central query window contains the following SQL script:

```
SELECT * FROM mov.Movie_Actor;

--d. List all the Rating information from the Rating table.
SELECT * FROM mov.Movie_Rating;

--e. List all the movie released in year "2010".
SELECT * FROM mov.Movies
WHERE Movie_Released_Year = 2010;

--f. List all the movie released by "The Weinstein Company" studio.
SELECT * FROM mov.Movies
WHERE Movie_Lead_Studio = 'The Weinstein Company';

--g. List all the movie released in "English".

--h. List all the movie whose name starts with 'G'.
```

The Results pane at the bottom shows the output of the query, displaying a table with the following data:

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	Cr
1	1000	Zack and Mini Make a Pomo	The Weinstein Company	English	Romance	101	41.94	Hollywood	100	20
2	1001	Youth in Revolt	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110	20
3	1009	My Week with Marilyn	The Weinstein Company	English	Drama	99	8.26	Hollywood	190	20

The status bar at the bottom indicates that the query was executed successfully, returning 3 rows in 00:00:00 seconds.

g. List all the movie released in “English”.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The main query window contains the following SQL script:

```
SELECT * FROM mov.Movie_Actor;

--d. List all the Rating information from the Rating table.
SELECT * FROM mov.Movie_Rating;

--e. List all the movie released in year "2010".
SELECT * FROM mov.Movies
WHERE Movie_Released_Year = 2010;

--f. List all the movie released by "The Weinstein Company" studio.
SELECT * FROM mov.Movies
WHERE Movie_Lead_Studio = 'The Weinstein Company';

--g. List all the movie released in "English".
SELECT * FROM mov.Movies
WHERE Movie_Language = 'English';

--h. List all the movie whose name starts with 'G'.
```

The Results pane at the bottom shows the output of the query for movies released in English. The status bar indicates 'Query executed successfully.' and '23 rows'.

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	Cn	
1	1000	Zack and Mini Make a Porno	2008	The Weinstein Company	English	Romance	101	41.94	Hollywood	100	20
2	1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110	20
3	1002	When in Rome	2010	Dianey	English	Comedy	91	43.04	Hollywood	120	20
4	1003	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130	20
5	1004	Water For Elephants	2011	20th Century Fox	English	Drama	120	117.09	Hollywood	140	20
6	1005	Watress	2007	Independent	English	Romance	108	22.18	Hollywood	150	20
7	1006	Twilight	2008	Summit	English	Romance	122	376.66	Hollywood	160	20
8	1007	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170	20
9	1008	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180	20
10	1009	My Week with Marilyn	2011	The Weinstein Company	English	Drama	99	8.26	Hollywood	190	20
11	1010	Music and Lyrics	2007	Warner Bros.	English	Romance	104	145.90	Hollywood	200	20
12	1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210	20
13	1012	Knocked Up	2007	Universal	English	Comedy	129	219.00	Hollywood	220	20
14	1013	Jane Eyre	2011	Universal	English	Romance	120	30.15	Hollywood	230	20
15	1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240	20
16	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250	20
17	1016	Gnomeo and Juliet	2011	Dianey	English	Animation	84	193.97	Hollywood	260	20
18	1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270	20
19	1018	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280	20
20	1019	Fireproof	2008	Independent	English	Drama	122	33.47	Hollywood	290	20
21	1020	Enchanted	2007	Dianey	English	Comedy	107	340.49	Hollywood	300	20
22	1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310	20
23	1022	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320	20

h. List all the movie whose name starts with 'G'.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the 'Movies' database structure. The 'Query Editor' in the center contains a SQL script with several queries, including the one for movies starting with 'G'. The 'Results' pane at the bottom shows the output of the executed query, displaying a table with 4 rows of movie data.

```
--a. List all the movie released in 2010.
SELECT * FROM mov.Movies
WHERE Movie_Released_Year = 2010;

--f. List all the movie released by "The Weinstein Company" studio.
SELECT * FROM mov.Movies
WHERE Movie_Lead_Studio = 'The Weinstein Company';

--g. List all the movie released in "English".
SELECT * FROM mov.Movies
WHERE Movie_Language = 'English';

--h. List all the movie whose name starts with 'G'.
SELECT * FROM mov.Movies
WHERE Movie_Name LIKE 'G%';

--i. Display all the movie under "Comedy" category.
```

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	CreatedOn	
1	1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240	2024-03-2
2	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250	2024-03-2
3	1016	Gnomeo and Juliet	2011	Disney	English	Animation	84	193.97	Hollywood	260	2024-03-2
4	1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270	2024-03-2

Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 4 rows

i. Display all the movie under “Comedy” category.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane contains a SQL query window with the following code:

```
--f. List all the movie released in "English".
SELECT * FROM mov.Movies
WHERE Movie_Language = 'English';

--g. List all the movie whose name starts with 'G'.
SELECT * FROM mov.Movies
WHERE Movie_Name LIKE 'G%';

--h. Display all the movie under "Comedy" category.
SELECT * FROM mov.Movies
WHERE Movie_Category = 'Comedy';

--i. Display all the movie where movie type is "Hollywood".
```

The bottom pane shows the 'Results' tab with a table of 15 rows. The table columns are: Movie_ID, Movie_Name, Movie_Released_Year, Movie_Lead_Studio, Movie_Language, Movie_Category, Movie_Duration_in_Min, Movie_Worldwide_Earning_in_\$M, Movie_Type, Director_ID, and Create. The data is as follows:

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	Create
1	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110	2024-
2	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120	2024-
3	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130	2024-
4	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170	2024-
5	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180	2024-
6	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210	2024-
7	Knocked Up	2007	Universal	English	Comedy	129	219.00	Hollywood	220	2024-
8	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240	2024-
9	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250	2024-
10	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270	2024-
11	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280	2024-
12	Enchanted	2007	Disney	English	Comedy	107	340.49	Hollywood	300	2024-
13	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320	2024-
14	3 Idiots	2009	Vinod Chopra Films	Hindi	Comedy	171	4000.00	Bollywood	330	2024-
15	Oh My God	2012		Hindi	Comedy	165	1200.00	Bollywood	390	2024-

The status bar at the bottom indicates 'Query executed successfully.' and shows the current position at line 277, column 1, with 15 rows of data.

j. Display all the movie where movie type is “Hollywood”.

The screenshot displays the Microsoft SQL Server Management Studio interface. The title bar indicates the connection to 'XPS\SQLSERVER2022.Movies (XPS\ammym (55))'. The Object Explorer on the left shows the database structure, including tables like 'mov.Movie_Actor', 'mov.Movie_Director', and 'mov.Movie_Rating'. The central query editor contains the following SQL script:

```
--i. Display all the movie under "Comedy" category.
SELECT * FROM mov.Movies
WHERE Movie_Category = 'Comedy';

--j. Display all the movie where movie type is "Hollywood".
SELECT * FROM mov.Movies
WHERE Movie_Type = 'Hollywood';

--k. Display all the "Female" directors.

--l. Display all the director whose Age is more than 45 years.

--m. Display all the Actors from "Los Angeles".
```

The 'Results' pane at the bottom shows the output of the query, displaying 23 rows of movie data. The status bar at the bottom indicates 'Query executed successfully.' and '23 rows'.

Movie_ID	Movie_Name	Movie_Released_Year	Movie_Lead_Studio	Movie_Language	Movie_Category	Movie_Duration_in_Min	Movie_Worldwide_Earning_in_\$M	Movie_Type	Director_ID	
1	1000	Zack and Mini Make a Pomo	2008	The Weinstein Company	English	Romance	101	41.94	Hollywood	100
2	1001	Youth in Revolt	2010	The Weinstein Company	English	Comedy	90	19.62	Hollywood	110
3	1002	When in Rome	2010	Disney	English	Comedy	91	43.04	Hollywood	120
4	1003	What Happens in Vegas	2008	Fox	English	Comedy	99	219.37	Hollywood	130
5	1004	Water For Elephants	2011	20th Century Fox	English	Drama	120	117.09	Hollywood	140
6	1005	Waitress	2007	Independent	English	Romance	108	22.18	Hollywood	150
7	1006	Twilight	2008	Summit	English	Romance	122	376.66	Hollywood	160
8	1007	Penelope	2008	Summit	English	Comedy	144	20.74	Hollywood	170
9	1008	Over Her Dead Body	2008	New Line	English	Comedy	95	20.71	Hollywood	180
10	1009	My Week with Marilyn	2011	The Weinstein Company	English	Drama	99	8.26	Hollywood	190
11	1010	Music and Lyrics	2007	Warner Bros.	English	Romance	104	145.90	Hollywood	200
12	1011	Leap Year	2010	Universal	English	Comedy	100	32.59	Hollywood	210
13	1012	Knocked Up	2007	Universal	English	Comedy	129	219.00	Hollywood	220
14	1013	Jane Eyre	2011	Universal	English	Romance	120	30.15	Hollywood	230
15	1014	Good Luck Chuck	2007	Lionsgate	English	Comedy	101	59.19	Hollywood	240
16	1015	Going the Distance	2010	Warner Bros.	English	Comedy	103	42.05	Hollywood	250
17	1016	Gnomeo and Juliet	2011	Disney	English	Animation	84	193.97	Hollywood	260
18	1017	Ghosts of Girlfriends Past	2009	Warner Bros.	English	Comedy	100	102.22	Hollywood	270
19	1018	Four Christmases	2008	Warner Bros.	English	Comedy	88	161.83	Hollywood	280
20	1019	Fireproof	2008	Independent	English	Drama	122	33.47	Hollywood	290
21	1020	Enchanted	2007	Disney	English	Comedy	107	340.49	Hollywood	300
22	1021	Dear John	2010	Sony	English	Drama	108	114.97	Hollywood	310
23	1022	Beginners	2011	Independent	English	Comedy	105	14.31	Hollywood	320

k. Display all the “Female” directors.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)', including 'Databases', 'System Databases', 'Database Snapshots', 'Movies', 'Database Diagrams', 'Tables', 'System Tables', 'FileTables', 'External Tables', 'Graph Tables', 'mov.Movie_Actor', 'mov.Movie_Director', 'mov.Movie_Rating', 'mov.Movies', and 'Dropped Ledger Tables'. The main query editor contains the following SQL script:

```
WHERE Movie_Name LIKE 'GX';

--i. Display all the movie under "Comedy" category.
SELECT * FROM mov.Movies
WHERE Movie_Category = 'Comedy';

--j. Display all the movie where movie type is "Hollywood".
SELECT * FROM mov.Movies
WHERE Movie_Type = 'Hollywood';

--k. Display all the "Female" directors.
SELECT * FROM mov.Movie_Director
WHERE Director_Gender = 'Female';

--l. Display all the director whose Age is more than 45 years.
```

The Results pane at the bottom shows the output of the query for female directors:

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn
1	Adrienne	Shelly	40	Female	2024-03-23
2	Nanette	Burstein	52	Female	2024-03-23

The status bar at the bottom indicates 'Query executed successfully.' and '2 rows'.

I. Display all the director whose Age is more than 45 years.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the 'Movies' database structure. The 'Query Editor' in the center contains a SQL query with several comments and a SELECT statement. The 'Results' pane at the bottom shows the output of the query, which is a table of directors with columns: Director_ID, Director_First_Name, Director_Last_Name, Director_Age_in_Years, Director_Gender, and CreatedOn. The status bar at the bottom indicates the query was executed successfully.

```
SELECT * FROM mov.Movies
WHERE Movie_Type = 'Hollywood';

--k. Display all the "Female" directors.
SELECT * FROM mov.Movie_Director
WHERE Director_Gender = 'Female';

--l. Display all the director whose Age is more than 45 years.
SELECT * FROM mov.Movie_Director
WHERE Director_Age_in_Years > 45;

--m. Display all the Actors from "Los Angeles".

--n. Display all the Actor whose Age is less than 50 years.
```

Director_ID	Director_First_Name	Director_Last_Name	Director_Age_in_Years	Director_Gender	CreatedOn
1	Kevin	Smith	52	Male	2024-03-23
2	Mark	Johnson	58	Male	2024-03-23
3	Francis	Lawrence	52	Male	2024-03-23
4	David	Slade	53	Male	2024-03-23
5	Mark	Palansky	53	Male	2024-03-23
6	Jeff	Lowell	49	Male	2024-03-23
7	Marc	Lawrence	95	Male	2024-03-23
8	Anand	Tucker	59	Male	2024-03-23
9	Judd	Apatow	55	Male	2024-03-23
10	Mark	Helfrich	49	Male	2024-03-23
11	Nanette	Burstein	52	Female	2024-03-23
12	Mark	Waters	58	Male	2024-03-23
13	Seth	Gordon	46	Male	2024-03-23
14	Alex	Kendrick	52	Male	2024-03-23
15	Kevin	Lima	60	Male	2024-03-23
16	Lasse	Hallström	76	Male	2024-03-23
17	Ewan	McGregor	52	Male	2024-03-23
18	Rajkumar	Hirani	60	Male	2024-03-23
19	Ashutosh	Gowariker	59	Male	2024-03-23
20	Karan	Johar	50	Male	2024-03-23
21	S.S	Rajamouli	49	Male	2024-03-23
22	Sukumar	NULL	53	Male	2024-03-23
23	Aditya	Chopra	51	Male	2024-03-23
24	Umesh	Shukla	52	Male	2024-03-23

m. Display all the Actors from “Los Angeles”.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'Movies Database - Final Project PT 2.sql'. The central query window contains the following SQL script:

```
--k. Display all the "Female" directors.
SELECT * FROM mov.Movie_Director
WHERE Director_Gender = 'Female';

--l. Display all the director whose Age is more than 45 years.
SELECT * FROM mov.Movie_Director
WHERE Director_Age_in_Years > 45;

--m. Display all the Actors from "Los Angeles".
SELECT * FROM mov.Movie_Actor
WHERE Actor_Location = 'Los Angeles';

--n. Display all the Actor whose Age is less than 50 years.
```

The 'Results' pane at the bottom shows the output of the third query, displaying 5 rows of actor data from Los Angeles:

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn
10	Seth	Rogen	53	Los Angeles	1000	2024-03-23
14	Robert	Pattinson	45	Los Angeles	1004	2024-03-23
16	Robert	Pattinson	45	Los Angeles	1006	2024-03-23
22	Judd	Apatow	58	Los Angeles	1012	2024-03-23
27	Matthew	McConaughey	52	Los Angeles	1017	2024-03-23

A status bar at the bottom indicates 'Query executed successfully.' and '5 rows'.

n. Display all the Actor whose Age is less than 50 years.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the 'Movies' database structure. The 'Query Editor' in the center contains a SQL query with several comments. The 'Results' pane at the bottom shows the output of the query, which is a table of actors with their details. A status bar at the bottom indicates the query was executed successfully.

```
SELECT * FROM mov.Movie_Actor
WHERE Actor_Location = 'Los Angeles';

--n. Display all the Actor whose Age is less than 50 years.
SELECT * FROM mov.Movie_Actor
WHERE Actor_Age_in_Years < 50;

--o. Display all the Actor whose name is starts from "J".

--p. Display all the Actor who is from "Los Angeles" or "New York".

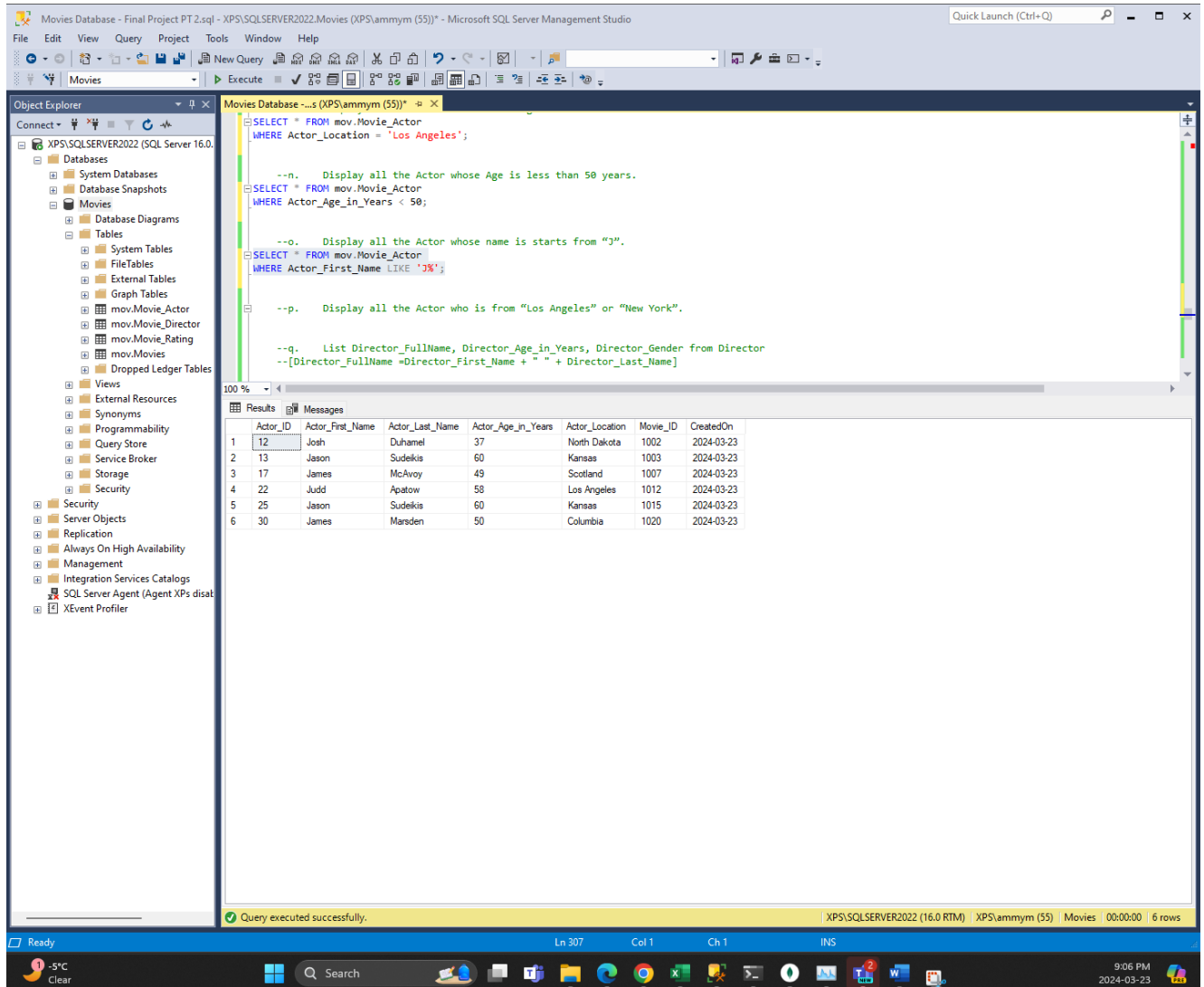
--q. List Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]

--r. List Director_FullName, Director_Age_in_Years, Director_Gender from Director whose Age is less than 45 years.
```

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn	
1	11	Michael	Cera	49	New York	1001	2024-03-23
2	12	Josh	Duhamel	37	North Dakota	1002	2024-03-23
3	14	Robert	Pattinson	45	Los Angeles	1004	2024-03-23
4	16	Robert	Pattinson	45	Los Angeles	1006	2024-03-23
5	17	James	McAvoy	49	Scotland	1007	2024-03-23
6	19	Kenneth	Branagh	44	Northem Ireland	1009	2024-03-23
7	21	Matthew	Goode	46	England	1011	2024-03-23
8	23	Michael	Fassbender	46	Germany	1013	2024-03-23
9	32	Mike	Mills	37	New York	1022	2024-03-23
10	37	Aliu	Ajun	49	India	1027	2024-03-23

Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\lammy (55) | Movies | 00:00:00 | 10 rows

o. Display all the Actor whose name is starts from “J”.



The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'Movies'. The central query editor contains the following SQL script:

```
--n. Display all the Actor whose Age is less than 50 years.
SELECT * FROM mov.Movie_Actor
WHERE Actor_Age_in_Years < 50;

--o. Display all the Actor whose name is starts from "J".
SELECT * FROM mov.Movie_Actor
WHERE Actor_First_Name LIKE 'J%';

--p. Display all the Actor who is from "Los Angeles" or "New York".

--q. List Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]
```

The 'Results' tab at the bottom shows the output of the query, displaying 6 rows of actor data:

	Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn
1	12	Josh	Duhamel	37	North Dakota	1002	2024-03-23
2	13	Jason	Sudekia	60	Kansas	1003	2024-03-23
3	17	James	McAvoy	49	Scotland	1007	2024-03-23
4	22	Judd	Apatow	58	Los Angeles	1012	2024-03-23
5	25	Jason	Sudekia	60	Kansas	1015	2024-03-23
6	30	James	Marsden	50	Columbia	1020	2024-03-23

The status bar at the bottom indicates 'Query executed successfully.' and shows the connection details: 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 6 rows'.

p. Display all the Actor who is from “Los Angeles” or “New York”.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)', including Databases, Tables, Views, and Security. The main query window shows a SQL script with several queries, with the last one selected:
`--p. Display all the Actor who is from "Los Angeles" or "New York".
SELECT * FROM mov.Movie_Actor
WHERE Actor_Location IN ('Los Angeles', 'New York');`
The Results pane at the bottom shows the output of this query as a table with 8 rows. The status bar at the bottom indicates 'Query executed successfully.' and '8 rows'.

Actor_ID	Actor_First_Name	Actor_Last_Name	Actor_Age_in_Years	Actor_Location	Movie_ID	CreatedOn
10	Seth	Rogen	53	Los Angeles	1000	2024-03-23
11	Michael	Cera	49	New York	1001	2024-03-23
14	Robert	Pattinson	45	Los Angeles	1004	2024-03-23
16	Robert	Pattinson	45	Los Angeles	1006	2024-03-23
18	Paul	Rudd	52	New York	1008	2024-03-23
22	Judd	Apatow	58	Los Angeles	1012	2024-03-23
27	Matthew	McConaughey	52	Los Angeles	1017	2024-03-23
32	Mike	Mills	37	New York	1022	2024-03-23

q. List Director_FullName, Director_Age_in_Years, Director_Gender from Director
 [Director_FullName =Director_First_Name + " " + Director_Last_Name]

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The central query window contains the following SQL code:

```
--p. Display all the Actor who is from "Los Angeles" or "New York".
SELECT * FROM mov.Movie_Actor
WHERE Actor_Location IN ('Los Angeles', 'New York');

--q. List Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]
SELECT Director_First_Name + ' ' + Director_Last_Name AS Director_FullName,
       Director_Age_in_Years,
       Director_Gender
FROM mov.Movie_Director;

--r. List Director_FullName, Director_Age_in_Years, Director_Gender from Director whose Age is less than 45 years.
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]
```

The Results pane at the bottom shows the output of the query, displaying 30 rows of data. The columns are Director_FullName, Director_Age_in_Years, and Director_Gender. The data is as follows:

Director_FullName	Director_Age_in_Years	Director_Gender
Kevin Smith	52	Male
Miguel Arteta	41	Male
Mark Johnson	58	Male
Tom Vaughan	37	Male
Francis Lawrence	52	Male
Adrienne Shelly	40	Female
David Slade	53	Male
Mark Palansky	53	Male
Jeff Lowell	49	Male
Simon Curtis	37	Male
Marc Lawrence	95	Male
Anand Tucker	59	Male
Judd Apatow	55	Male
Cary Fukunaga	45	Male
Mark Helfrich	49	Male
Nanette Burstein	52	Female
James McAvoy	44	Male
Mark Waters	58	Male
Seth Gordon	46	Male
Alex Kendrick	52	Male
Kevin Lima	60	Male
Lasse Hallström	76	Male
Ewan McGregor	52	Male
Rajkumar Hirani	60	Male
Ashutosh Gowariker	59	Male
Karan Johar	50	Male
S.S. Rajamouli	49	Male
NULL	53	Male
Aditya Chopra	51	Male
Imesh Shukla	52	Male

The status bar at the bottom indicates the query was executed successfully on the 'XPS\SQLSERVER2022 (16.0 RTM)' instance, returning 30 rows.

r. List Director_FullName, Director_Age_in_Years, Director_Gender from Director whose Age is less than 45 years. [Director_FullName=Director_First_Name + " " + Director_Last_Name]

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'Movies Database'. The main query window contains two SQL queries. The first query lists all directors with their full names, ages, and genders. The second query filters the results to show only directors who are 45 years old or younger. The results pane at the bottom shows the output of the second query, displaying 5 rows of data.

```
--q. List Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName=Director_First_Name + " " + Director_Last_Name]
SELECT Director_First_Name + ' ' + Director_Last_Name AS Director_FullName,
       Director_Age_in_Years,
       Director_Gender
FROM mov.Movie_Director;

--r. List Director_FullName, Director_Age_in_Years, Director_Gender from Director whose Age is less than 45 years.
--[Director_FullName=Director_First_Name + " " + Director_Last_Name]
SELECT Director_First_Name + ' ' + Director_Last_Name AS Director_FullName,
       Director_Age_in_Years,
       Director_Gender
FROM mov.Movie_Director
WHERE Director_Age_in_Years <= 45;
```

Director_FullName	Director_Age_in_Years	Director_Gender
Miguel Arteta	41	Male
Tom Vaughan	37	Male
Adrienne Shelly	40	Female
Simon Curtis	37	Male
James McAvoy	44	Male

Query executed successfully.

6. Write the following Query based on the above datasets.

a. Display all the Movies and their Actors information based on the relationship.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the 'Movies Database' structure, including tables like 'mov.Movie_Director', 'mov.Movie_Actor', and 'mov.Movie_Rating'. The 'Query Editor' in the center contains a SQL query that joins 'mov.Movie_Actor' and 'mov.Movie' tables. The 'Results' pane at the bottom shows the output of the query, which is a list of movies and their actors.

```
Director_Age_in_Years,
Director_Gender
FROM mov.Movie_Director
WHERE Director_Age_in_Years < 45;

--8. Write the following Query based on the above datasets.
--a. Display all the Movies and their Actors information based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID;

--b. Display the Movies name and their Ratings.
--c. Display all the Movies, Actors, and Directors information based on the relationship.
--d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.
--e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.
--f. Display all the Movies information whose Rating_Rotten_Tomatoes is more than 90%.
--9. Write the following Query based on the above datasets.
```

Movie_Name	Actor_First_Name	Actor_Last_Name
1 Zack and Mit Make a Porno	Seth	Rogen
2 Youth in Revolt	Michael	Cera
3 When in Rome	Josh	Duhamel
4 What Happens in Vegas	Jason	Sudekis
5 Water For Elephants	Robert	Pattinson
6 Watress	Nathan	Fillion
7 Twilight	Robert	Pattinson
8 Penelope	James	McAvoy
9 Over Her Dead Body	Paul	Rudd
10 My Week with Marilyn	Kenneth	Branagh
11 Music and Lyrics	Hugh	Grant
12 Leap Year	Matthew	Goode
13 Knocked Up	Judd	Apatow
14 Jane Eyre	Michael	Fassabender
15 Good Luck Chuck	Dane	Cook
16 Going the Distance	Jason	Sudekis
17 Gnomeo and Juliet	Kelly	Asbury
18 Ghosts of Girlfriends Past	Matthew	McConaughey
19 Four Christmases	Vince	Vaughn
20 Fireproof	Kirk	Cameron
21 Enchanted	James	Marsden
22 Dear John	Channing	Tatum
23 Beginners	Mike	Mills
24 3 Idiots	Aamir	Khan
25 Lagaan	Aamir	Khan
26 My Name is Khan	Shah Rukh	Khan
27 Baahubali	Prabhas	NULL
28 Dilwale Dulhania Le Jayenge	Allu	Ajun
29 Oh My God	Shah Rukh	Khan
30 Pushpa	Akshay	Kumar

Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 30 rows

b. Display the Movies name and their Ratings.

The screenshot displays the Microsoft SQL Server Management Studio interface. The 'Object Explorer' on the left shows the 'Movies' database structure, including tables like 'mov.Movie_Director', 'mov.Movie_Rating', and 'mov.Movies'. The 'Query Editor' in the center contains the following SQL script:

```
Director_Age_in_Years,  
Director_Gender  
FROM mov.Movie_Director  
WHERE Director_Age_in_Years < 45;  
  
--b. Write the following Query based on the above datasets.  
--a. Display all the Movies and their Actors information based on the relationship.  
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name  
FROM mov.Movies m  
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID;  
  
--b. Display the Movies name and their Ratings.  
SELECT m.Movie_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes  
FROM mov.Movies m  
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;  
  
--c. Display all the Movies, Actors, and Directors information based on the relationship.  
--d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.
```

The 'Results' pane at the bottom shows the output of the query, displaying a table with 30 rows of movie data:

Movie_Name	Rating_Audience_Score	Rating_Rotten_Tomatoes
1 Zack and Mit Make a Porno	70	64
2 Youth in Revolt	52	68
3 When in Rome	44	15
4 What Happens in Vegas	72	28
5 Water For Elephants	72	60
6 Waitress	67	89
7 Twilight	82	49
8 Penelope	74	52
9 Over Her Dead Body	47	15
10 My Week with Marilyn	84	83
11 Music and Lyrics	70	63
12 Leap Year	49	21
13 Knocked Up	83	91
14 Jane Eyre	77	85
15 Good Luck Chuck	61	3
16 Going the Distance	56	53
17 Gnomeo and Juliet	52	56
18 Ghosts of Girlfriends Past	47	27
19 Four Christmases	52	26
20 Fireproof	51	40
21 Enchanted	80	93
22 Dear John	66	29
23 Beginners	80	84
24 3 Idiots	95	100
25 Lagaan	81	95
26 My Name Is Khan	79	83
27 Baahubali	80	90
28 Dilwale Dulhania Le Jayenge	76	82
29 Oh My God	85	100
30 Pishua	81	74

The status bar at the bottom indicates 'Query executed successfully.' and '30 rows'.

c. Display all the Movies, Actors, and Directors information based on the relationship.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a SQL query window with the following queries:

```
--b. Display the Movies name and their Ratings.
SELECT m.Movie_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--c. Display all the Movies, Actors, and Directors information based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID;

--d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.
--e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.
--f. Display all the Movies information whose Rating_Rotten_Tomatoes is more than 90%.

--9. Write the following Query based on the above datasets.
--a. Create new table MovieCopy and copy all records of Movie table.
--b. Create a new table DirectorCopy and copy only the schema of director table.
--c. Create new table ActorCopy and copy all records of Actor table.
```

The bottom pane shows the 'Results' tab with the following data:

Movie_Name	Actor_First_Name	Actor_Last_Name	Director_First_Name	Director_Last_Name
1 Zack and Mit Make a Porno	Seth	Rogen	Kevin	Smith
2 Youth in Revolt	Michael	Cera	Miguel	Arteta
3 When in Rome	Josh	Duhamel	Mark	Johnson
4 What Happens in Vegas	Jason	Sudekis	Tom	Vaughan
5 Water For Elephants	Robert	Pattinson	Francis	Lawrence
6 Waitress	Nathan	Fillion	Adrienne	Shelly
7 Twilight	Robert	Pattinson	David	Slade
8 Penelope	James	McAvoy	Mark	Palansky
9 Over Her Dead Body	Paul	Rudd	Jeff	Lowell
10 My Week with Marilyn	Kenneth	Branagh	Simon	Curtis
11 Music and Lyrics	Hugh	Grant	Marc	Lawrence
12 Leap Year	Matthew	Goode	Anand	Tucker
13 Knocked Up	Judd	Apatow	Judd	Apatow
14 Jane Eyre	Michael	Fassbender	Cary	Fukunaga
15 Good Luck Chuck	Dane	Cook	Mark	Helfrich
16 Going the Distance	Jason	Sudekis	Nanette	Burstein
17 Gnomeo and Juliet	Kelly	Aisbury	James	McAvoy
18 Ghosts of Girlfriends Past	Matthew	McConaughey	Mark	Waters
19 Four Christmases	Vince	Vaughn	Seth	Gordon
20 Fireproof	Kirk	Cameron	Alex	Kendrick
21 Enchanted	James	Marsden	Kevin	Lima
22 Dear John	Channing	Tatum	Lasse	Hallström
23 Beginners	Mike	Mills	Ewan	McGregor
24 3 Idiots	Aamir	Khan	Rajkumar	Hirani
25 Lagaan	Aamir	Khan	Ashutosh	Gowariker
26 My Name Is Khan	Shah Rukh	Khan	Karan	Johar
27 Baahubali	Prabhas	NULL	S.S	Rajamouli
28 Dilwale Dulhania Le Jayenge	Allu	Ajun	Aditya	Chopra
29 Oh My God	Shah Rukh	Khan	Umesh	Shukla
30 Puthna	Akshay	Kumar	Sukumar	Nilli

The status bar at the bottom indicates 'Query executed successfully.' and '30 rows'.

d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The main window shows a query with four parts: (b) displaying movie names and ratings, (c) displaying movies, actors, and directors, (d) displaying movies, actors, directors, and ratings, and (e) displaying the same information for movies with a rating above 80%. The query results are displayed in a table with 7 columns: Movie_Name, Actor_First_Name, Actor_Last_Name, Director_First_Name, Director_Last_Name, Rating_Audience_Score, and Rating_Rotten_Tomatoes. The results show 30 rows of data, including movies like 'Zack and Mit Make a Porno', 'Youth in Revolt', 'When in Rome', etc.

```
--b. Display the Movies name and their Ratings.
SELECT m.Movie_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--c. Display all the Movies, Actors, and Directors information based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID;

--d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.
```

	Movie_Name	Actor_First_Name	Actor_Last_Name	Director_First_Name	Director_Last_Name	Rating_Audience_Score	Rating_Rotten_Tomatoes
1	Zack and Mit Make a Porno	Seth	Rogen	Kevin	Smith	70	64
2	Youth in Revolt	Michael	Cera	Miguel	Arteta	52	68
3	When in Rome	Josh	Duhamel	Mark	Johnson	44	15
4	What Happens in Vegas	Jason	Sudekis	Tom	Vaughan	72	28
5	Water For Elephants	Robert	Pattinson	Francis	Lawrence	72	60
6	Waltress	Nathan	Fillon	Adrienne	Shelly	67	89
7	Twilight	Robert	Pattinson	David	Slade	82	49
8	Penelope	James	McAvoy	Mark	Palensky	74	52
9	Over Her Dead Body	Paul	Rudd	Jeff	Lowell	47	15
10	My Week with Marilyn	Kenneth	Branagh	Simon	Curtis	84	83
11	Music and Lyrics	Hugh	Grant	Marc	Lawrence	70	63
12	Leap Year	Matthew	Goode	Anand	Tucker	49	21
13	Knocked Up	Judd	Apatow	Judd	Apatow	83	91
14	Jane Eyre	Michael	Fassbender	Cary	Fukunaga	77	85
15	Good Luck Chuck	Dane	Cook	Mark	Helfrich	61	3
16	Going the Distance	Jason	Sudekis	Nanette	Burstein	56	53
17	Gnomeo and Juliet	Kelly	Asbury	James	McAvoy	52	56
18	Ghosts of Girlfriends Past	Matthew	McConaughey	Mark	Waters	47	27
19	Four Christmases	Vince	Vaughn	Seth	Gordon	52	26
20	Fireproof	Kirk	Cameron	Alex	Kendrick	51	40
21	Enchanted	James	Marsden	Kevin	Lima	80	93
22	Dear John	Channing	Tatum	Lasse	Hallström	66	29
23	Beginners	Mike	Mills	Ewan	McGregor	80	84
24	3 Idiots	Aamir	Khan	Rajkumar	Hirani	95	100
25	Lagaan	Aamir	Khan	Ashutosh	Gowariker	81	95
26	My Name Is Khan	Shah Rukh	Khan	Karan	Johar	79	83
27	Baahubali	Prabhas	NULL	S S	Rajamouli	80	90
28	Divale Dulhania Le Jayenge	Allu	Ajun	Aditya	Chopra	76	82
29	Oh My God	Shah Rukh	Khan	Umesh	Shukla	85	100
30	Puthna	Akshay	Kumar	Sikumar	NULL	81	74

e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The main query window contains the following SQL code:

```
--d. Display all the Movies, Actors, Directors, and Movie Rating information based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID
WHERE r.Rating_Audience_Score > 80;

--f. Display all the Movies information whose Rating_Rotten_Tomatoes is more than 90%.

--9. Write the following Query based on the above datasets.
--a. Create new table MovieCopy and copy all records of Movie table.
--b. Create a new table DirectorCopy and copy only the schema of director table.
```

The Results pane shows the output of the first query, displaying 7 rows of movie information:

	Movie_Name	Actor_First_Name	Actor_Last_Name	Director_First_Name	Director_Last_Name	Rating_Audience_Score	Rating_Rotten_Tomatoes
1	Twilight	Robert	Pattinson	David	Slade	82	49
2	My Week with Marilyn	Kenneth	Branagh	Simon	Curtis	84	83
3	Knocked Up	Judd	Apatow	Judd	Apatow	83	91
4	3 Idiots	Aamir	Khan	Rajkumar	Hirani	95	100
5	Lagaan	Aamir	Khan	Ashutosh	Gowankar	81	95
6	Oh My God	Shah Rukh	Khan	Umesh	Shukla	85	100
7	Pushpa	Akshay	Kumar	Sukumar	NULL	81	74

The status bar at the bottom indicates 'Query executed successfully.' and '7 rows'.

f. Display all the Movies information whose Rating_Rotten_Tomatoes is more than 90%.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)', including 'Databases', 'Tables', 'Views', and 'Security'. The central query editor contains the following SQL code:

```
--e. Display all the Movies, Actors, Directors, and Movie Rating information whose Rating_Audience_Score is more than 80% based on the relationship.
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, d.Director_First_Name, d.Director_Last_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Director d ON m.Movie_ID = d.Director_ID
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID
WHERE r.Rating_Audience_Score > 80;

--f. Display all the Movies information whose Rating_Rotten_Tomatoes is more than 90%.
SELECT m.Movie_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID
WHERE r.Rating_Rotten_Tomatoes > 90;

--9. Write the following Query based on the above datasets.
--a. Create new table MovieCopy and copy all records of Movie table.
--b. Create a new table DirectorCopy and copy only the schema of director table.
--c. Create new table ActorCopy and copy all records of Actor table.
--d. Create new table RatingCopy and copy all records of Rating table.
```

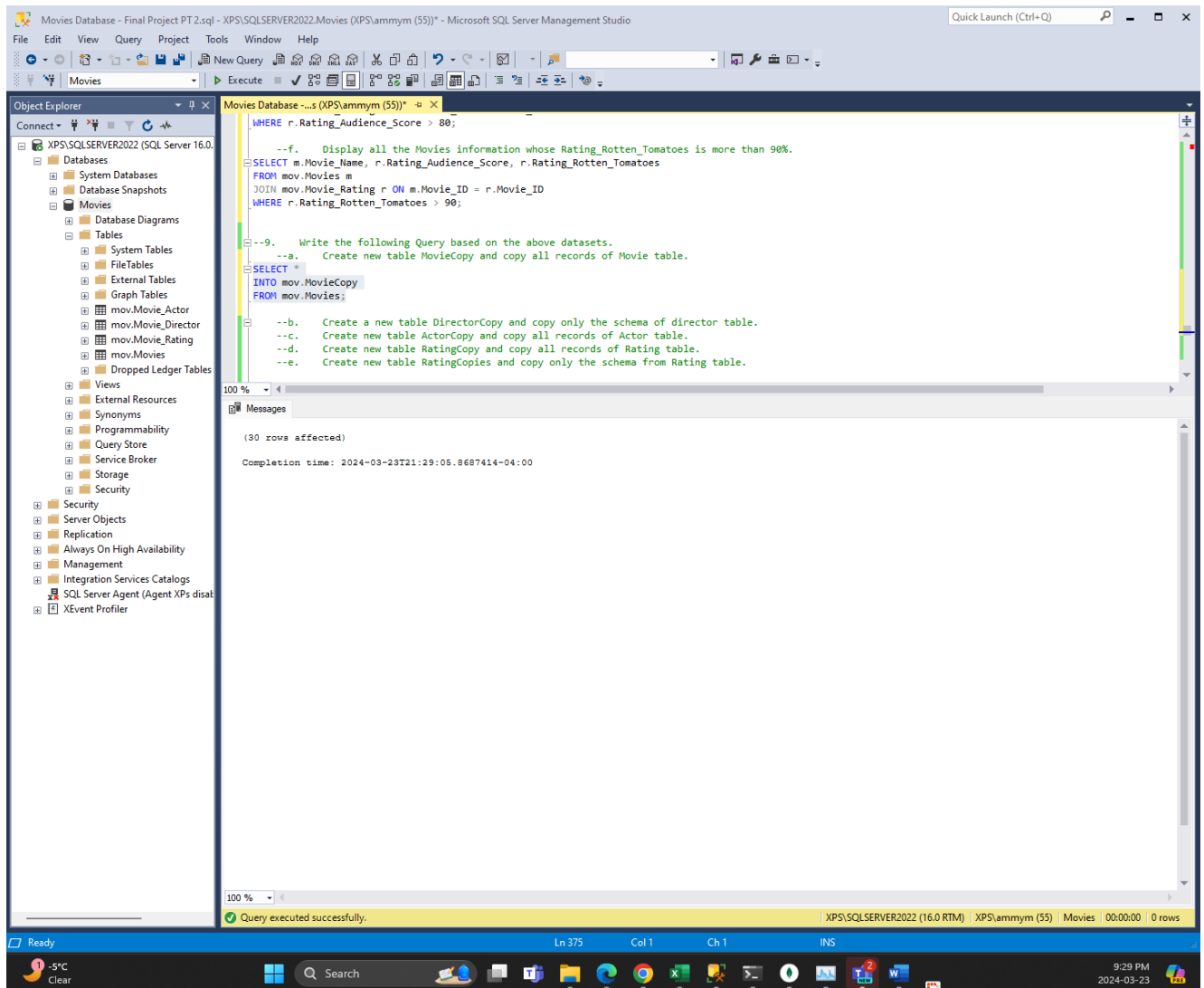
The Results pane at the bottom shows the output of the second query, displaying 5 rows of movie information:

	Movie_Name	Rating_Audience_Score	Rating_Rotten_Tomatoes
1	Knocked Up	83	91
2	Enchanted	80	93
3	3 Idiots	95	100
4	Lagaan	81	95
5	Oh My God	85	100

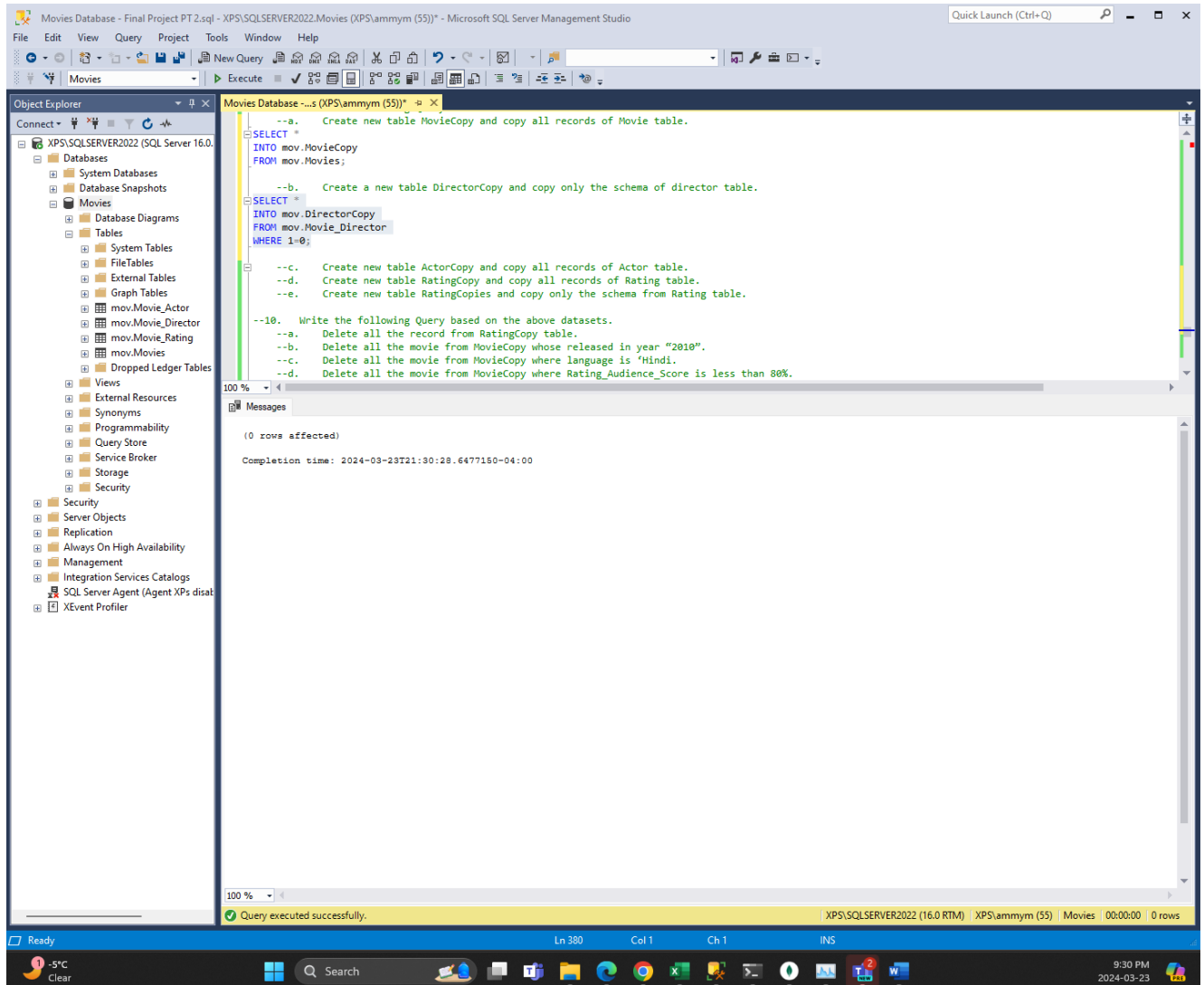
The status bar at the bottom indicates 'Query executed successfully.' and '5 rows'.

7. Write the following Query based on the above datasets.

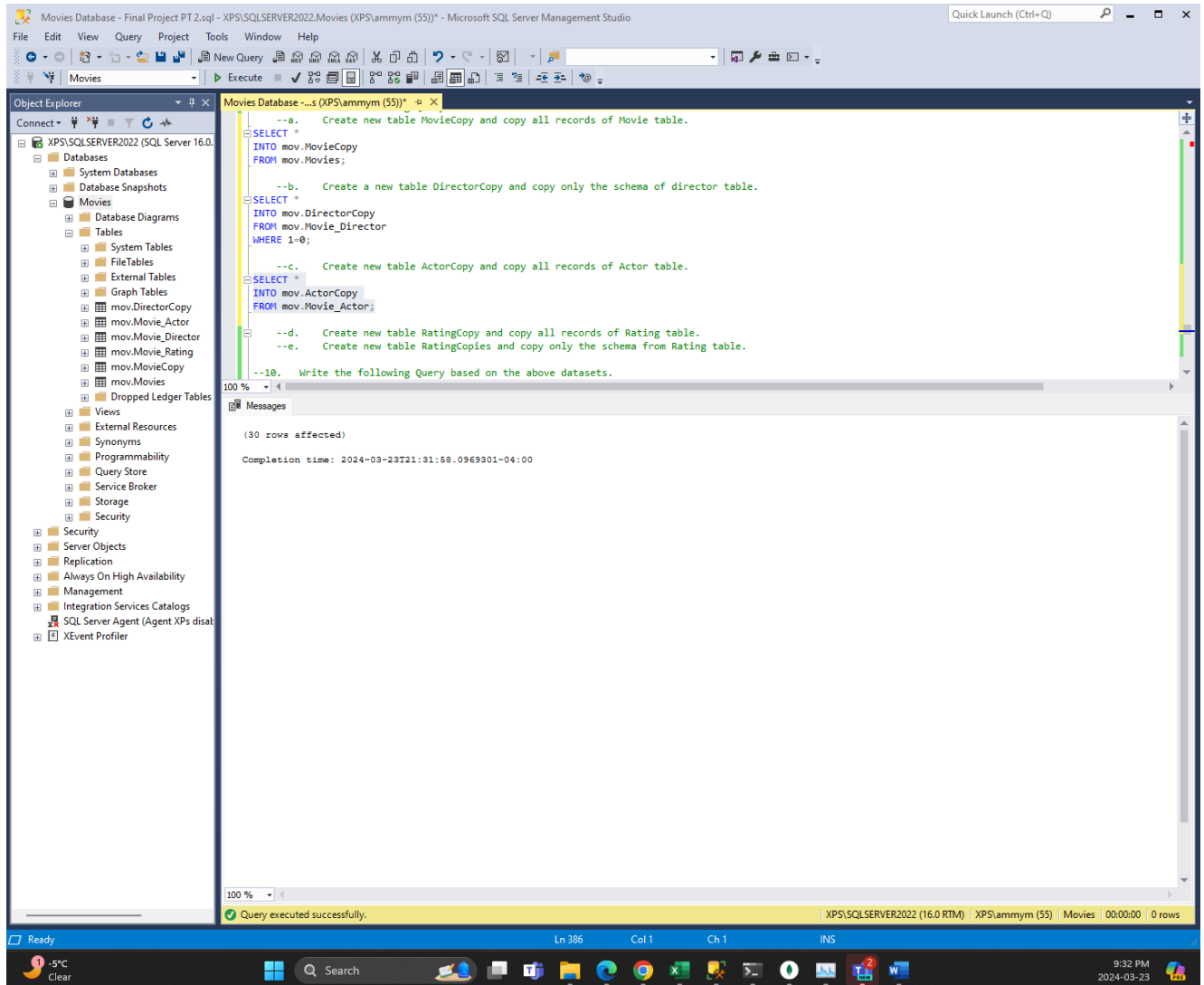
a. Create new table MovieCopy and copy all records of Movie table.



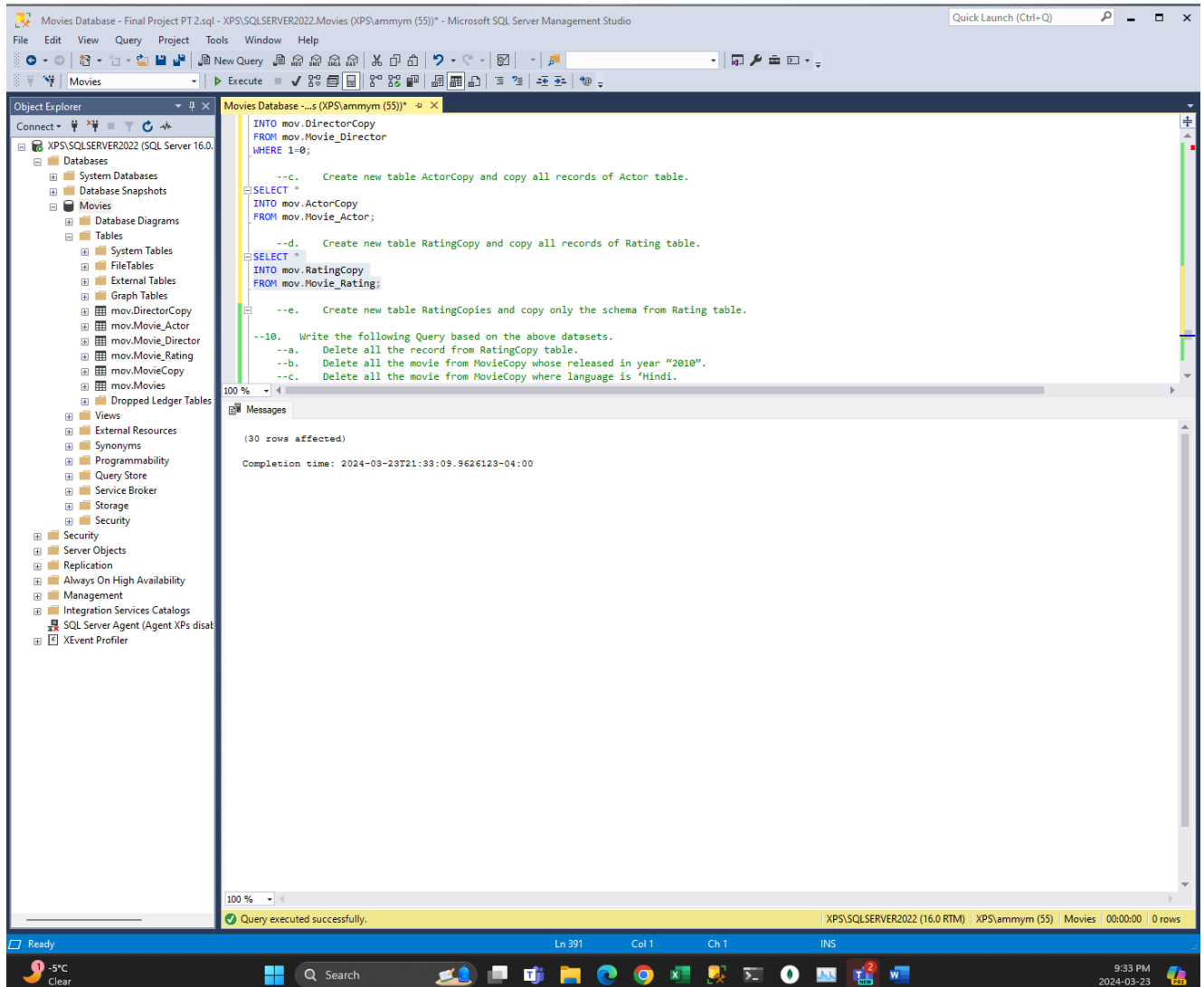
b. Create a new table DirectorCopy and copy only the schema of director table.



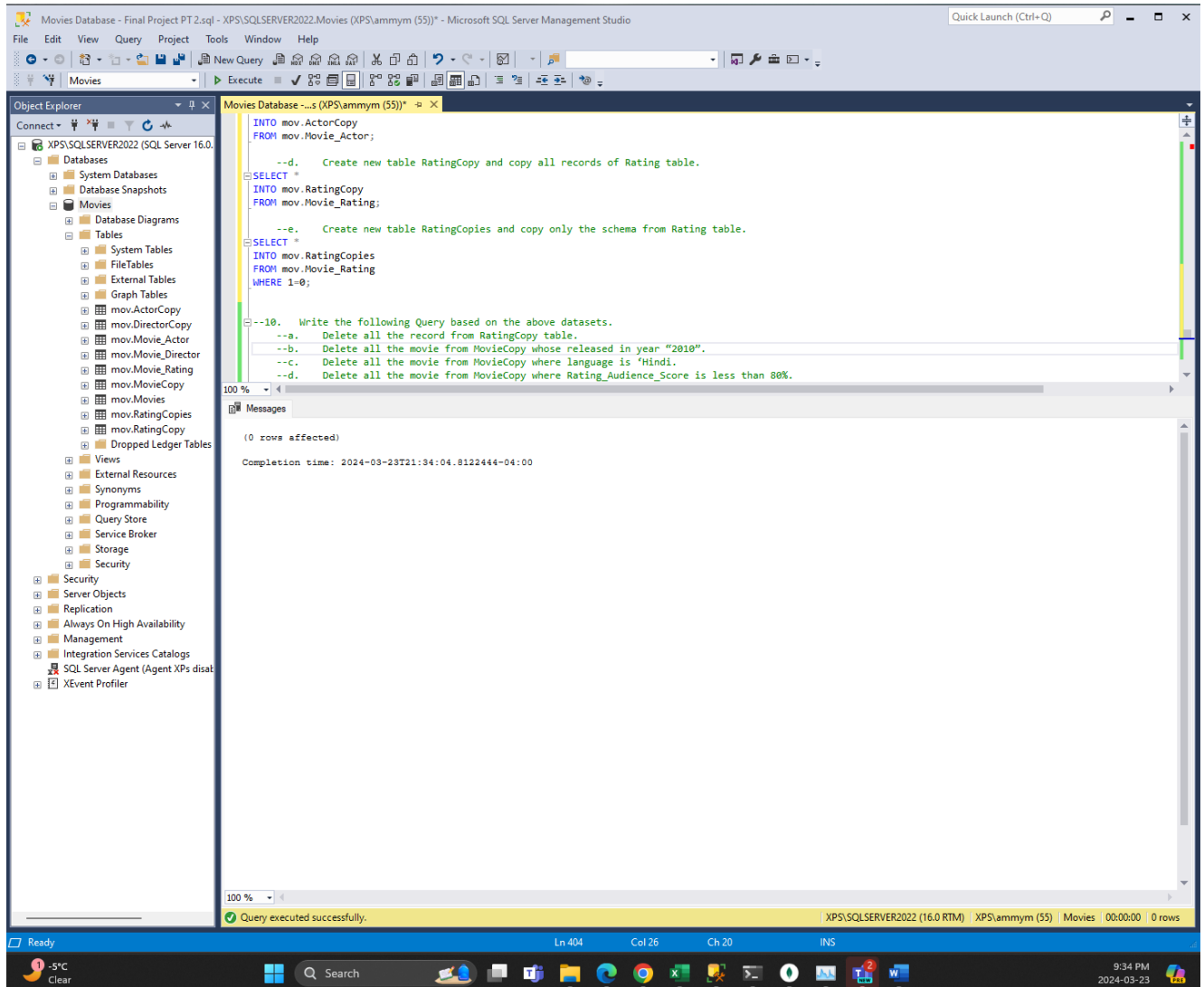
c. Create new table ActorCopy and copy all records of Actor table.



d. Create new table RatingCopy and copy all records of Rating table.

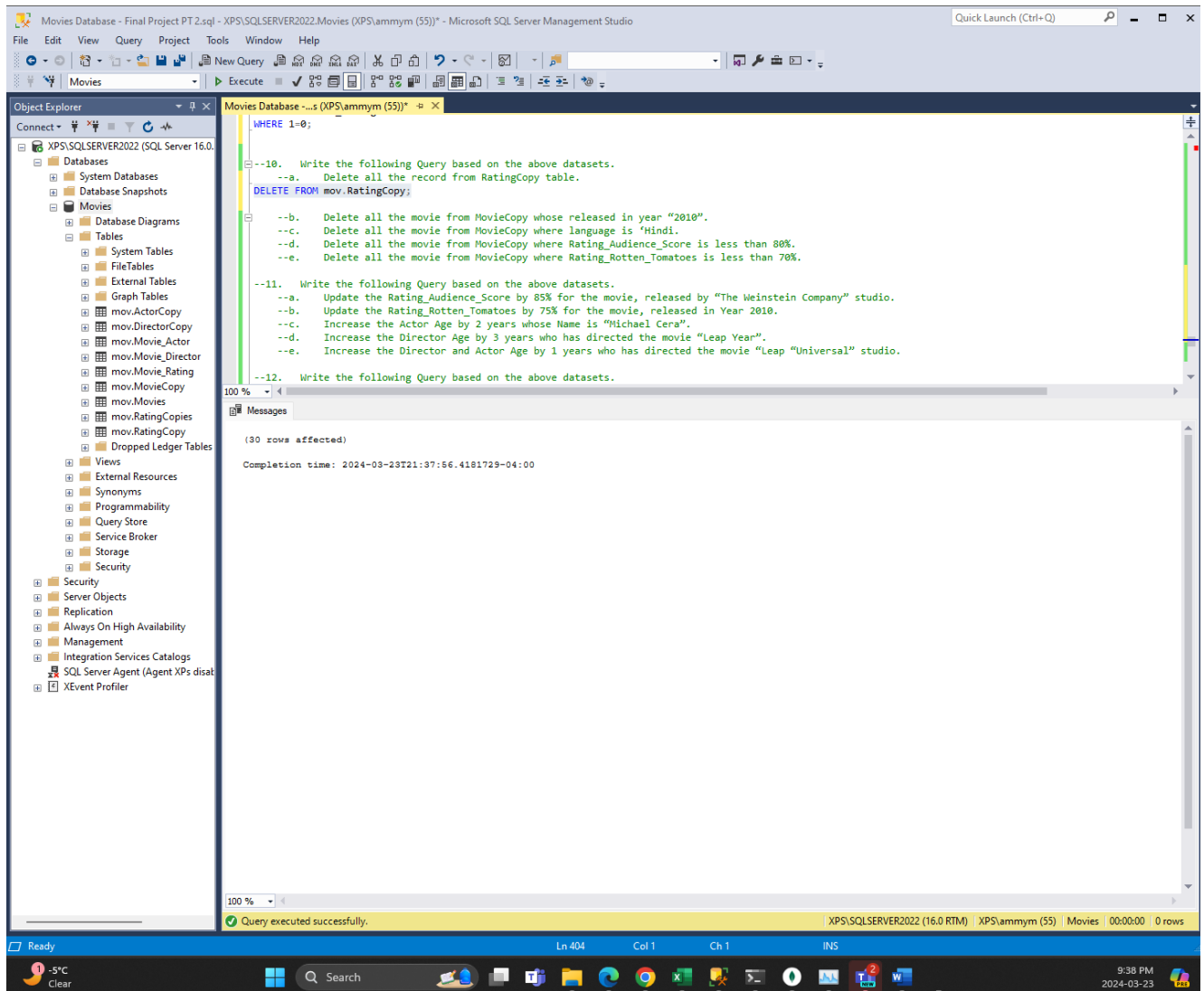


e. Create new table RatingCopies and copy only the schema from Rating table.

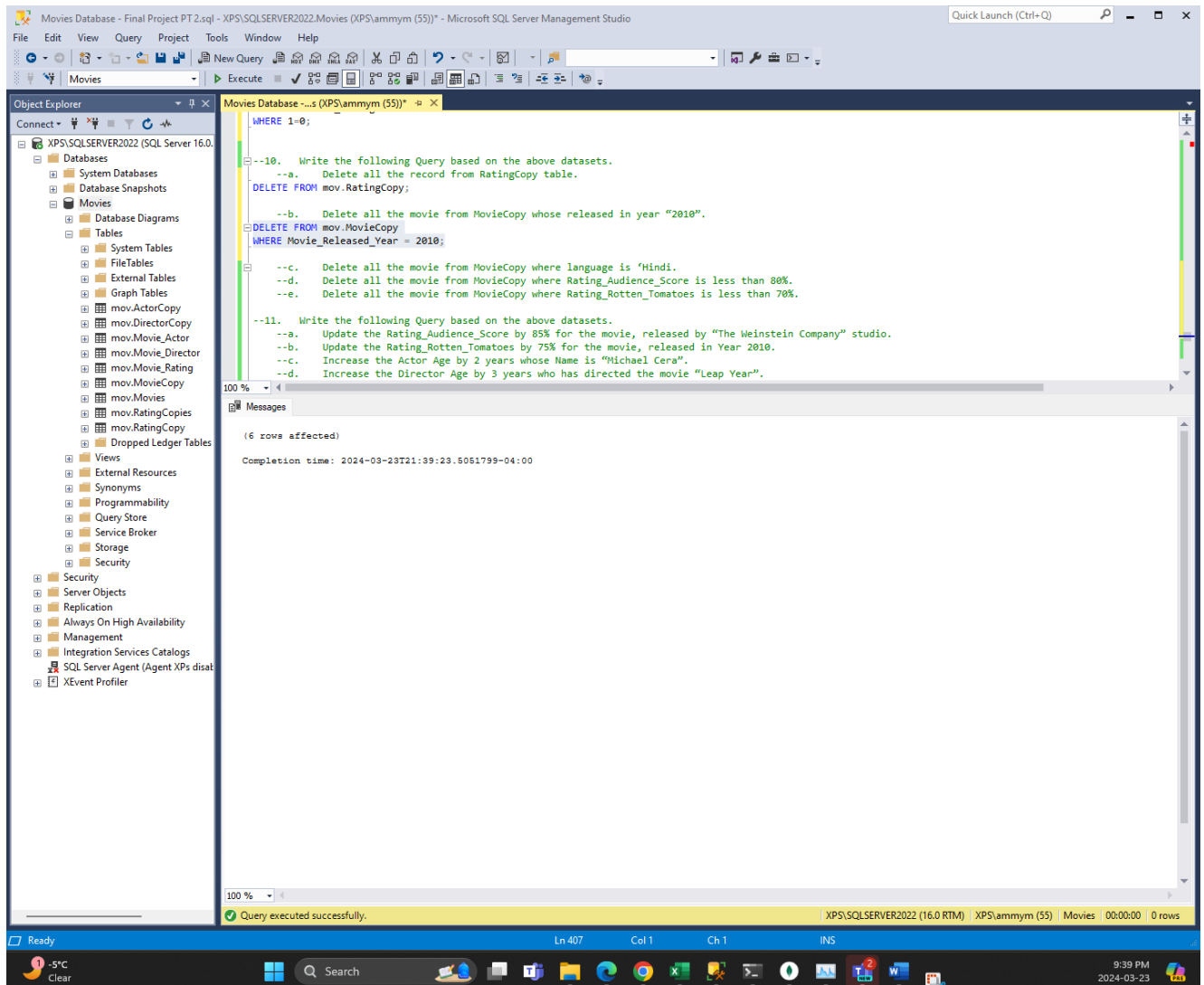


8. Write the following Query based on the above datasets.

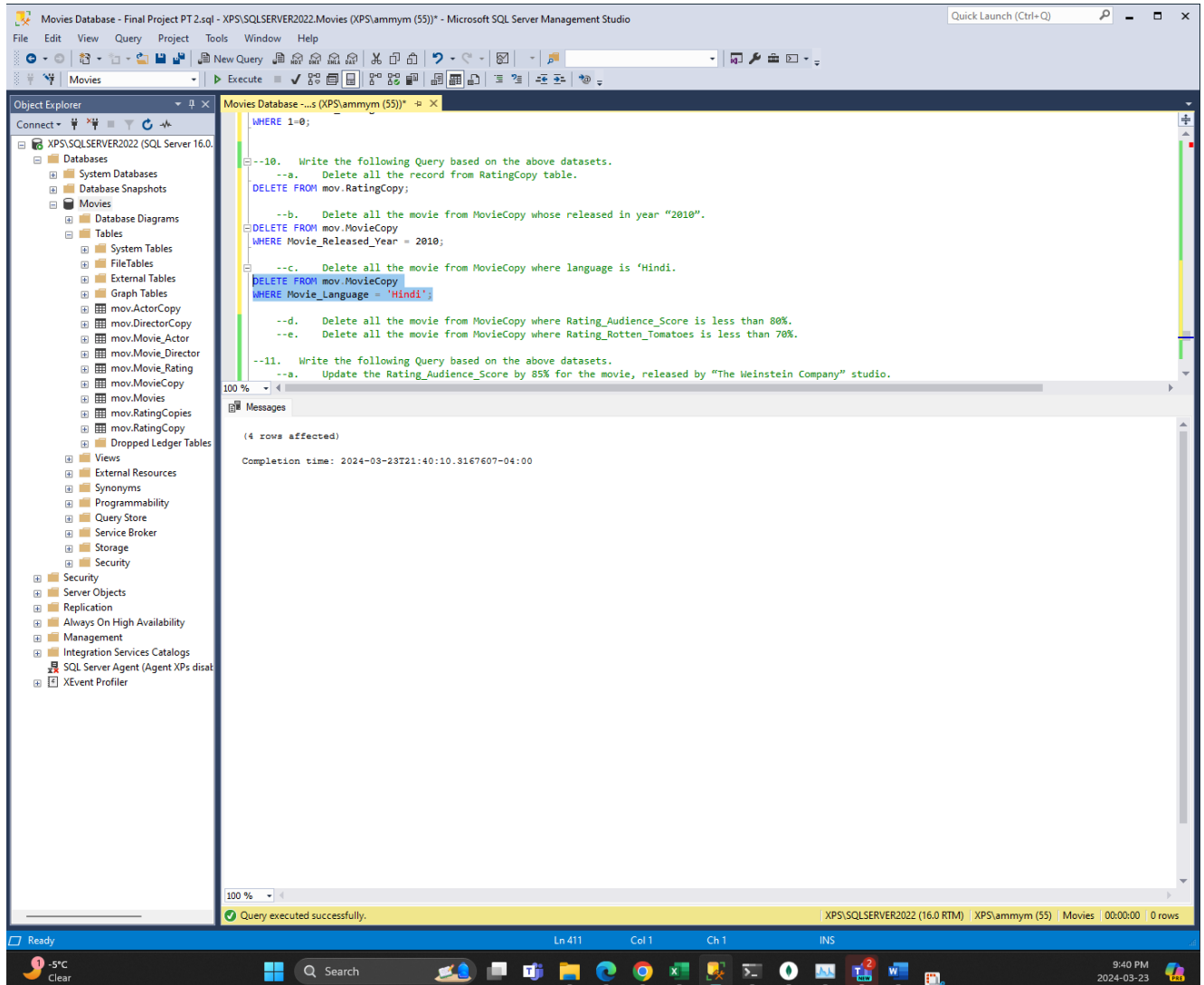
a. Delete all the record from RatingCopy table.



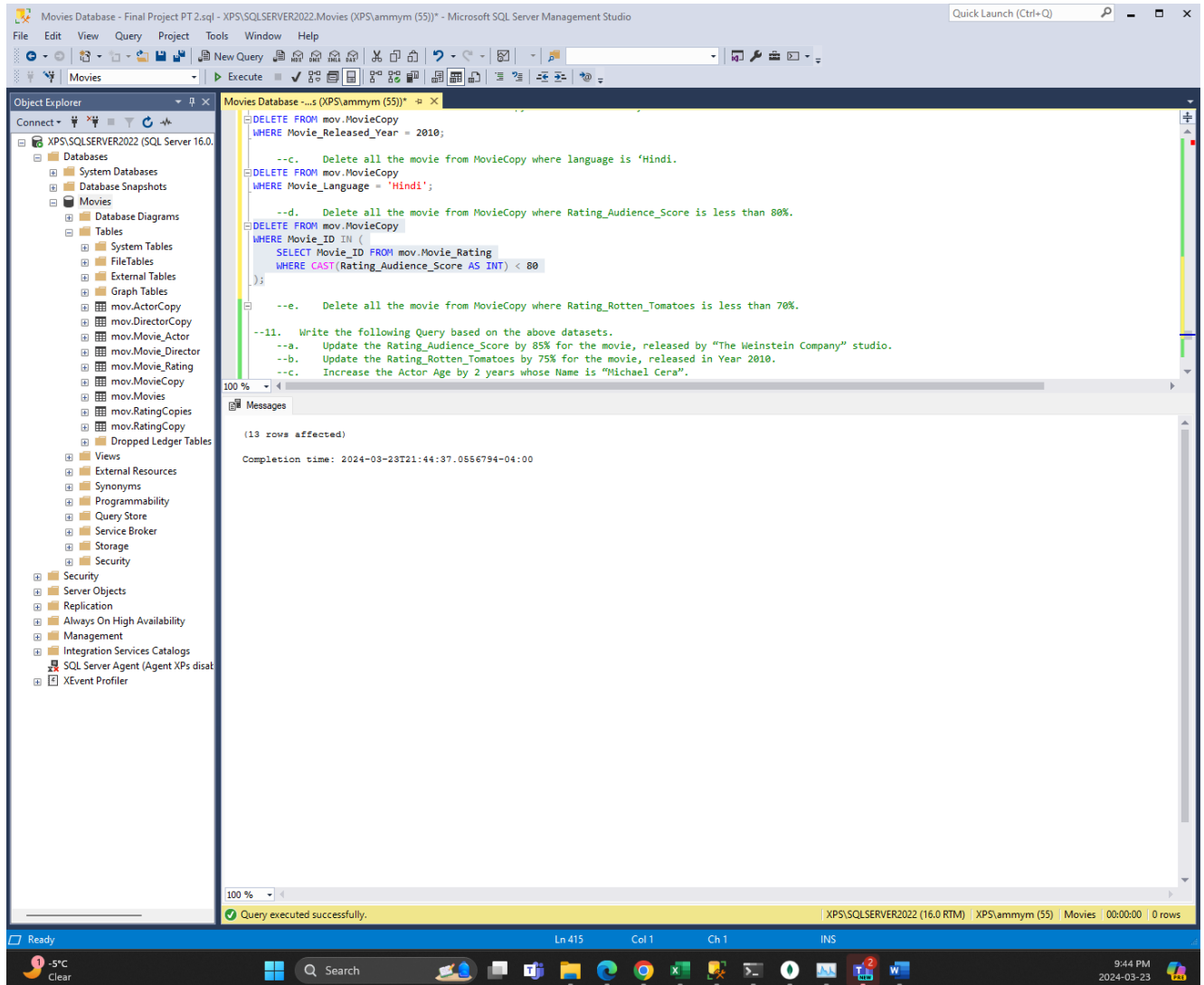
b. Delete all the movie from MovieCopy whose released in year “2010”.



c. Delete all the movie from MovieCopy where language is 'Hindi'.



d. Delete all the movie from MovieCopy where Rating_Audience_Score is less than 80%.



e. Delete all the movie from MovieCopy where Rating_Rotten_Tomatoes is less than 70%.

The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The title bar indicates the connection to 'Movies Database - Final Project PT 2.sql - XPS\SQLSERVER2022\Movies (XPS\ammym (55))'. The Object Explorer on the left shows the database structure, including tables like 'mov.MovieCopy', 'mov.Movie_Rating', and 'mov.RatingCopies'. The main query window contains the following SQL script:

```
--d. Delete all the movie from MovieCopy where Rating_Audience_Score is less than 80%.
DELETE FROM mov.MovieCopy
WHERE Movie_Language = 'Hindi';

--d. Delete all the movie from MovieCopy where Rating_Audience_Score is less than 80%.
DELETE FROM mov.MovieCopy
WHERE Movie_ID IN (
    SELECT Movie_ID FROM mov.Movie_Rating
    WHERE CAST(Rating_Audience_Score AS INT) < 80
);

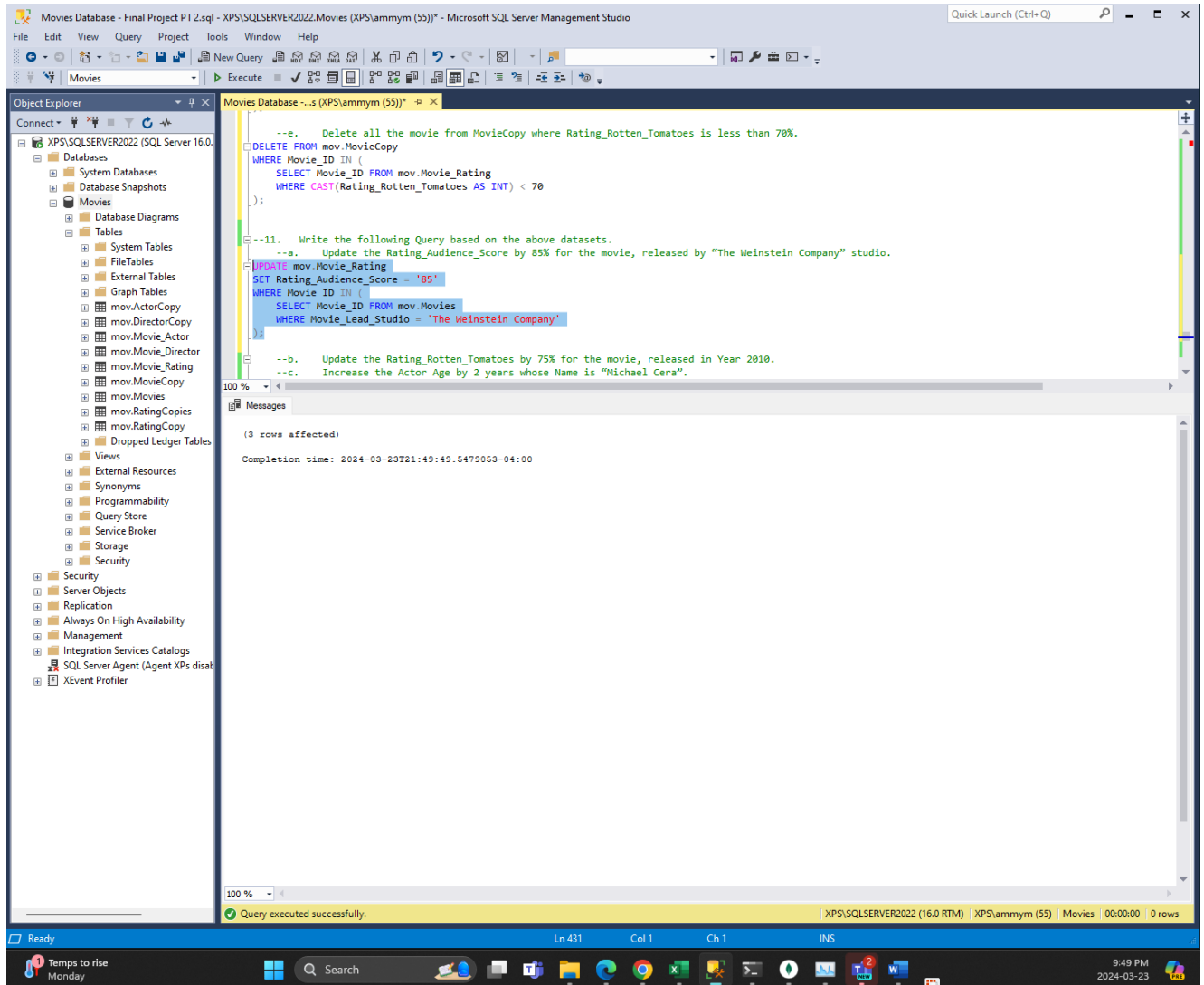
--e. Delete all the movie from MovieCopy where Rating_Rotten_Tomatoes is less than 70%.
DELETE FROM mov.MovieCopy
WHERE Movie_ID IN (
    SELECT Movie_ID FROM mov.Movie_Rating
    WHERE CAST(Rating_Rotten_Tomatoes AS INT) < 70
);

--11. Write the following Query based on the above datasets.
--a. Update the Rating_Audience_Score by 85% for the movie, released by "The Weinstein Company" studio.
```

The Messages pane at the bottom shows the execution results: '(1 row affected)' and 'Completion time: 2024-03-23T21:45:23.7826938-04:00'. A status bar at the bottom indicates 'Query executed successfully.' and shows the current position in the script (Ln 422, Col 1, Ch 1, INS).

9. Write the following Query based on the above datasets.

a. Update the Rating_Audience_Score by 85% for the movie, released by “The Weinstein Company” studio.



b. Update the Rating_Rotten_Tomatoes by 75% for the movie, released in Year 2010.

The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The title bar indicates the connection to 'Movies Database - Final Project PT 2.sql - XPS\SQLSERVER2022\Movies (XPS\ammym (55))'. The left-hand 'Object Explorer' pane shows the database structure, including tables like 'mov.Movie_Rating'. The central 'Query Editor' pane contains the following SQL script:

```
WHERE Movie_Lead_Studio = 'The Weinstein Company';

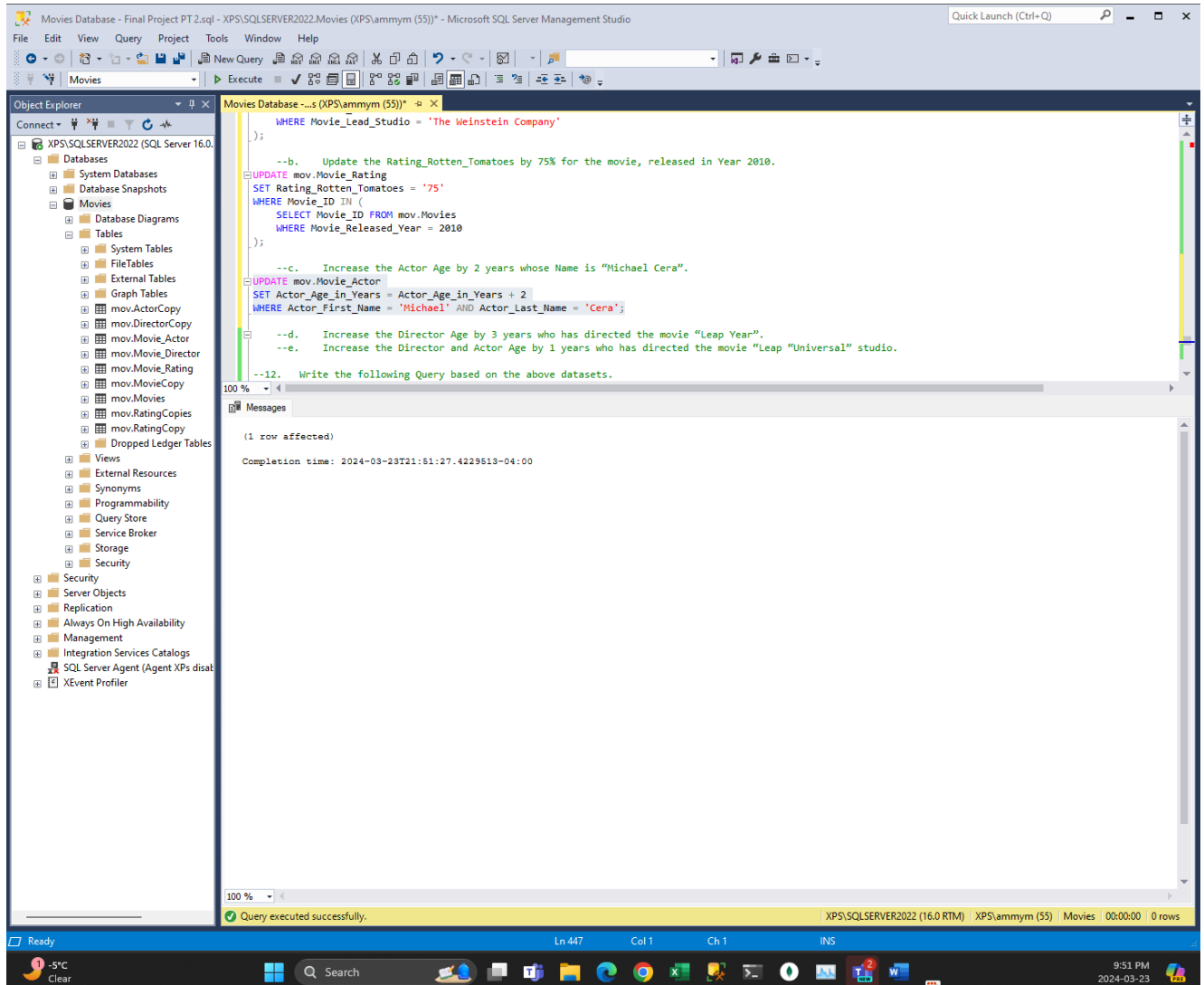
--b. Update the Rating_Rotten_Tomatoes by 75% for the movie, released in Year 2010.
UPDATE mov.Movie_Rating
SET Rating_Rotten_Tomatoes = '75'
WHERE Movie_ID IN (
    SELECT Movie_ID FROM mov.Movies
    WHERE Movie_Released_Year = 2010
);

--c. Increase the Actor Age by 2 years whose Name is "Michael Cera".
--d. Increase the Director Age by 3 years who has directed the movie "Leap Year".
--e. Increase the Director and Actor Age by 1 years who has directed the movie "Leap "Universal" studio.

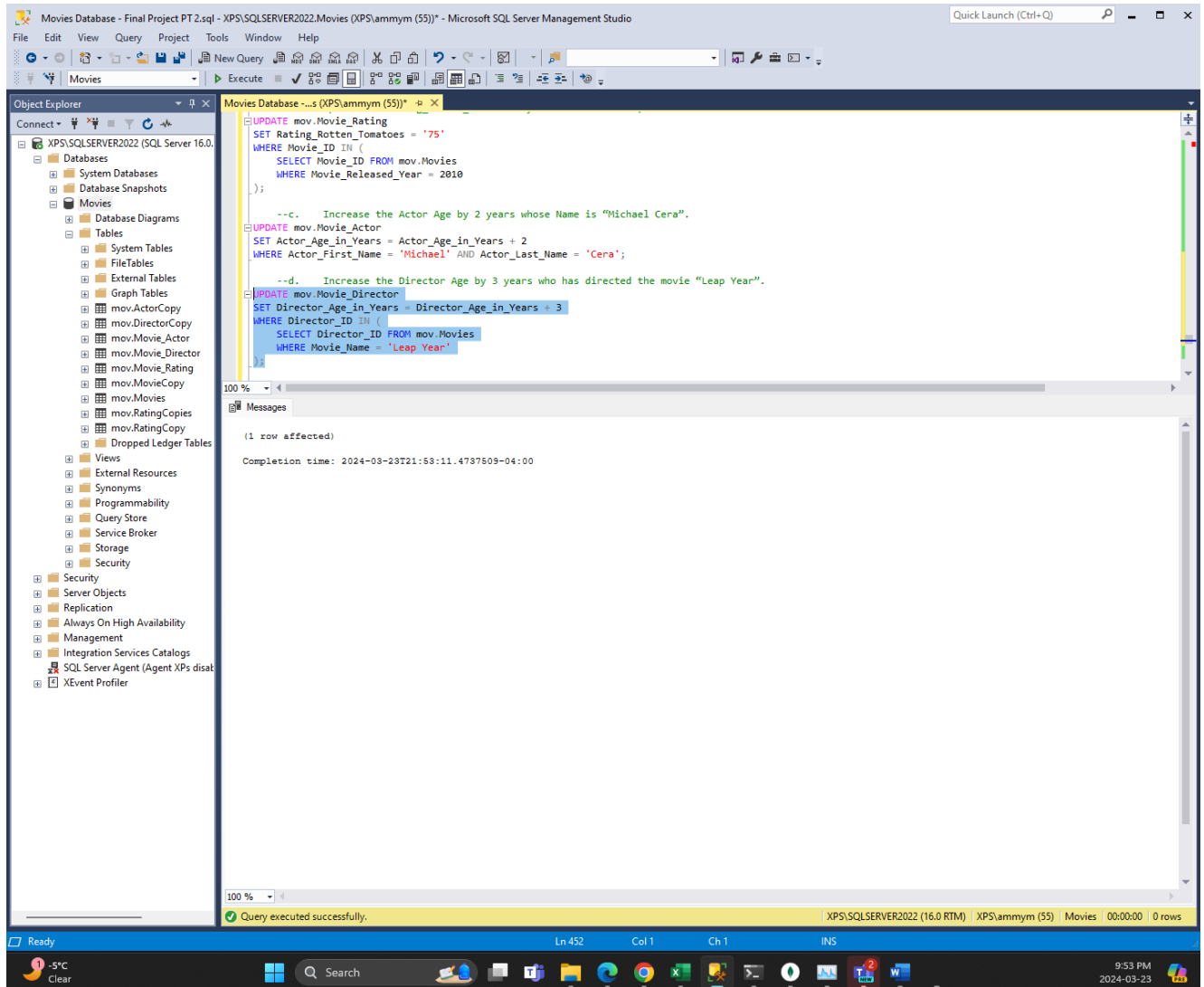
--12. Write the following Query based on the above datasets.
--a. Create a view to display all the movie information's.
--b. Create a view to display all the movies and their rating information.
--c. Create a view to display all the movies and their actor information.
--d. Create a view to display all the movies, rating, actor along with director information.
```

The 'Messages' pane at the bottom shows the execution results: '(6 rows affected)' and 'Completion time: 2024-03-23T21:50:27.0637888-04:00'. A status bar at the bottom of the window confirms 'Query executed successfully.' and shows the current position at 'Ln 439, Col 1, Ch 1, INS'.

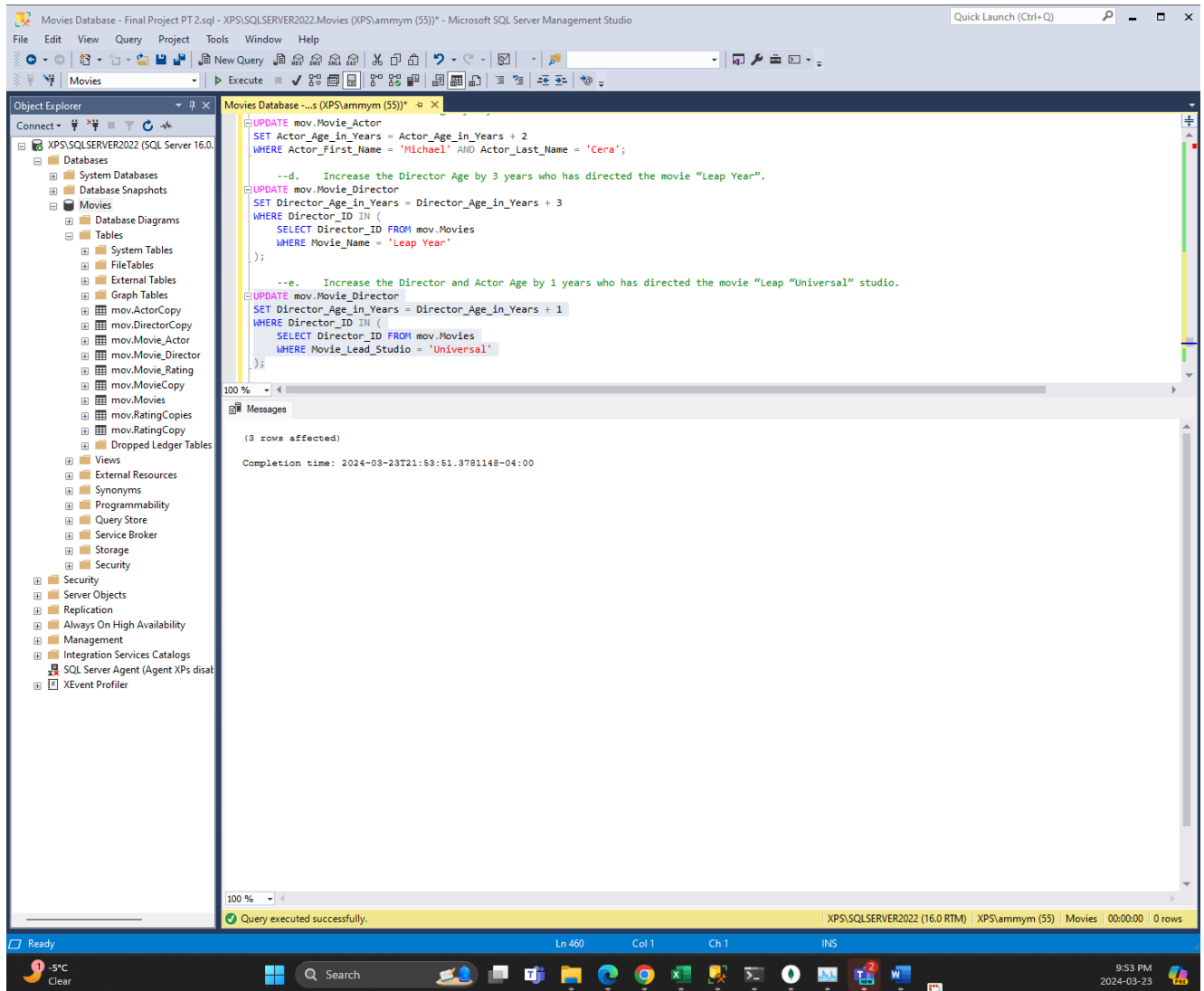
c. Increase the Actor Age by 2 years whose Name is “Michael Cera”.

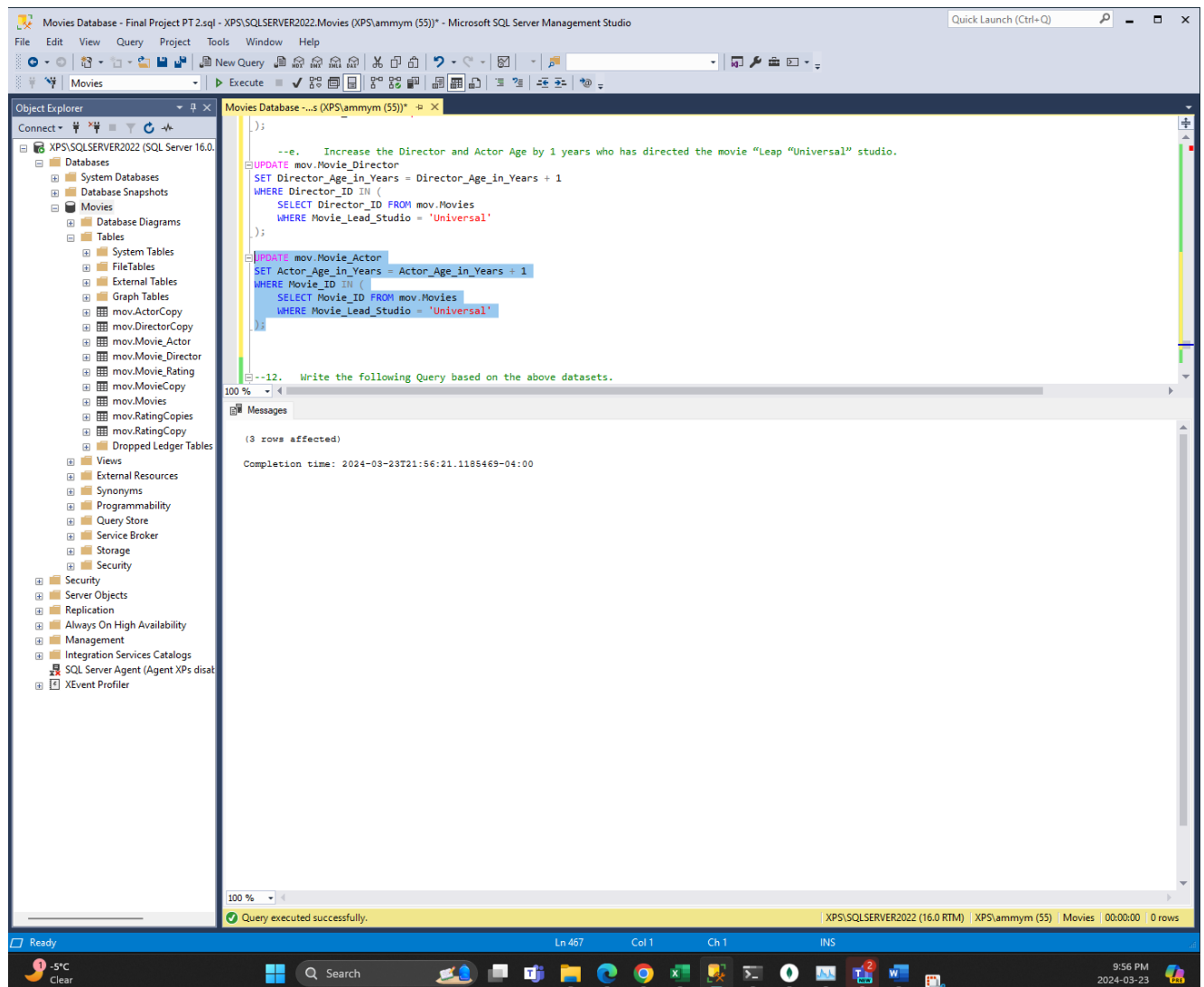


d. Increase the Director Age by 3 years who has directed the movie “Leap Year”.

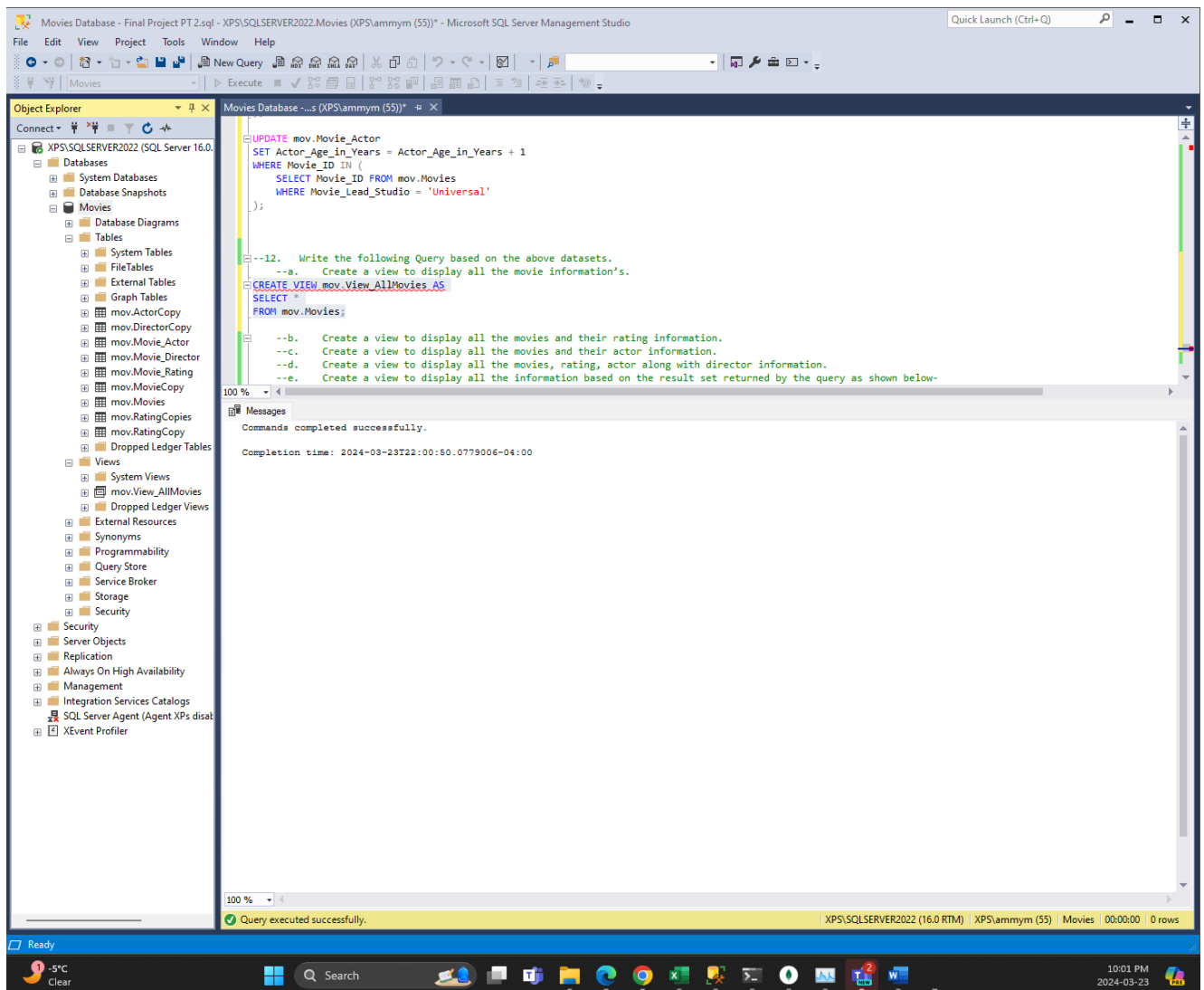


e. Increase the Director and Actor Age by 1 years who has directed the movie “Leap “Universal” studio.

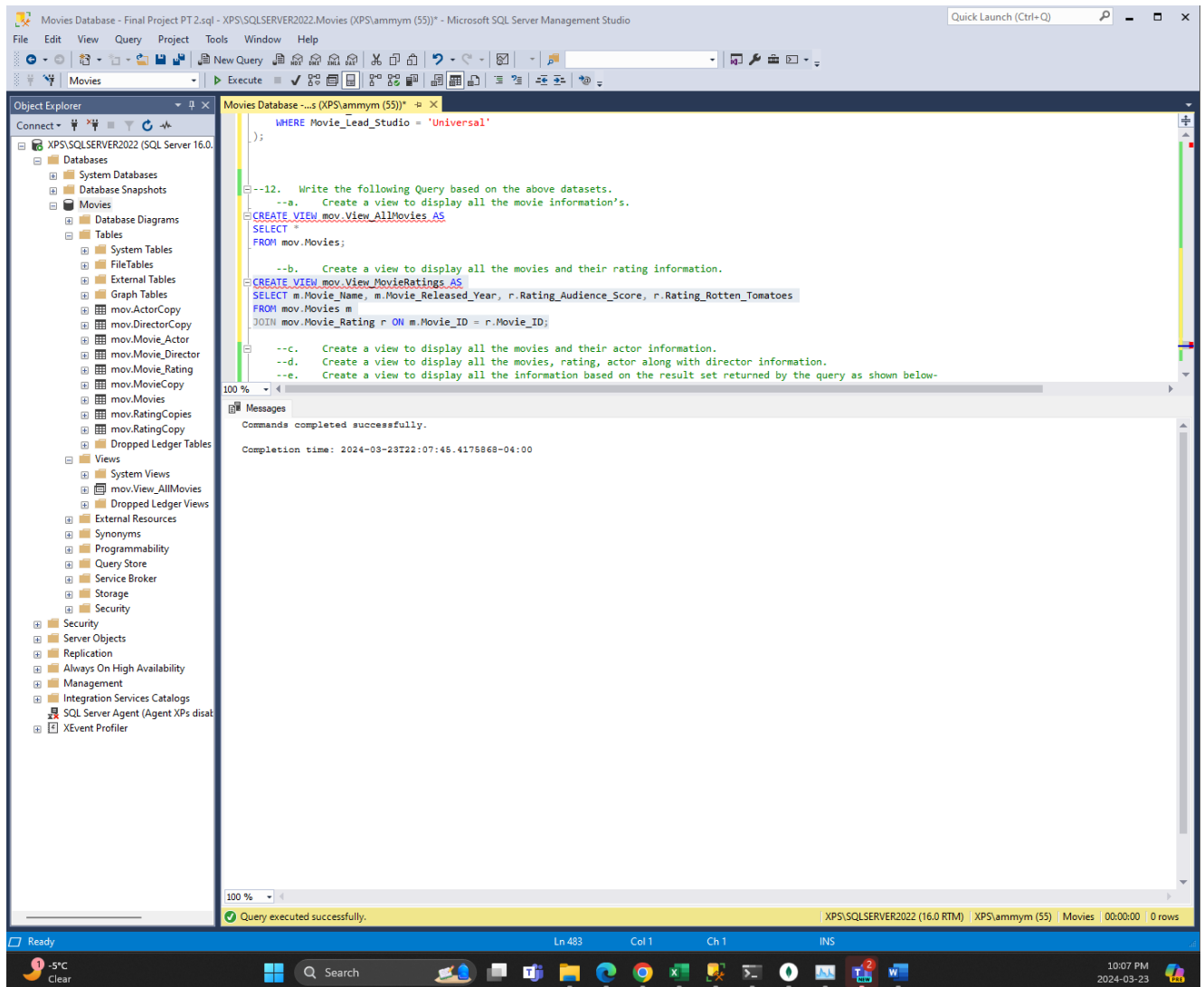




10. Write the following Query based on the above datasets.
- a. Create a view to display all the movie information's.



b. Create a view to display all the movies and their rating information.



c. Create a view to display all the movies and their actor information.

The screenshot displays the Microsoft SQL Server Management Studio interface. The title bar indicates the connection to 'XPS\SQLSERVER2022 (XPS\ammym (55))'. The Object Explorer on the left shows the database structure, including tables like 'mov.Movie_Rating' and 'mov.Movie_Actor'. The central query editor contains the following SQL script:

```
--b. Create a view to display all the movies and their rating information.
CREATE VIEW mov.View_MovieRatings_AS
SELECT m.Movie_Name, m.Movie_Released_Year, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--c. Create a view to display all the movies and their actor information.
CREATE VIEW mov.View_MovieActors_AS
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, a.Actor_Age_in_Years, a.Actor_Location
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID;

--d. Create a view to display all the movies, rating, actor along with director information.
--e. Create a view to display all the information based on the result set returned by the query as shown below-
--List movie, Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]
```

The Messages pane at the bottom shows a successful execution: 'Commands completed successfully.' and 'Completion time: 2024-03-23T22:09:16.6421305-04:00'. The status bar at the bottom indicates 'Query executed successfully.' and shows the current position at line 489, column 1.

d. Create a view to display all the movies, rating, actor along with director information.

The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a SQL query window with the following code:

```
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--c. Create a view to display all the movies and their actor information.
CREATE VIEW mov.View_MovieActors AS
SELECT m.Movie_Name, a.Actor_First_Name, a.Actor_Last_Name, a.Actor_Age_in_Years, a.Actor_Location
FROM mov.Movies m
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID;

--d. Create a view to display all the movies, rating, actor along with director information.
CREATE VIEW mov.View_AllInfo AS
SELECT m.Movie_Name, d.Director_First_Name + ' ' + d.Director_Last_Name AS Director_FullName,
a.Actor_First_Name, a.Actor_Last_Name, r.Rating_Audience_Score, r.Rating_Rotten_Tomatoes
FROM mov.Movies m
JOIN mov.Movie_Director d ON m.Director_ID = d.Director_ID
JOIN mov.Movie_Actor a ON m.Movie_ID = a.Movie_ID
JOIN mov.Movie_Rating r ON m.Movie_ID = r.Movie_ID;

--e. Create a view to display all the information based on the result set returned by the query as shown below
--List movie, Director_FullName, Director_Age_in_Years, Director_Gender from Director
--[Director_FullName =Director_First_Name + " " + Director_Last_Name]
```

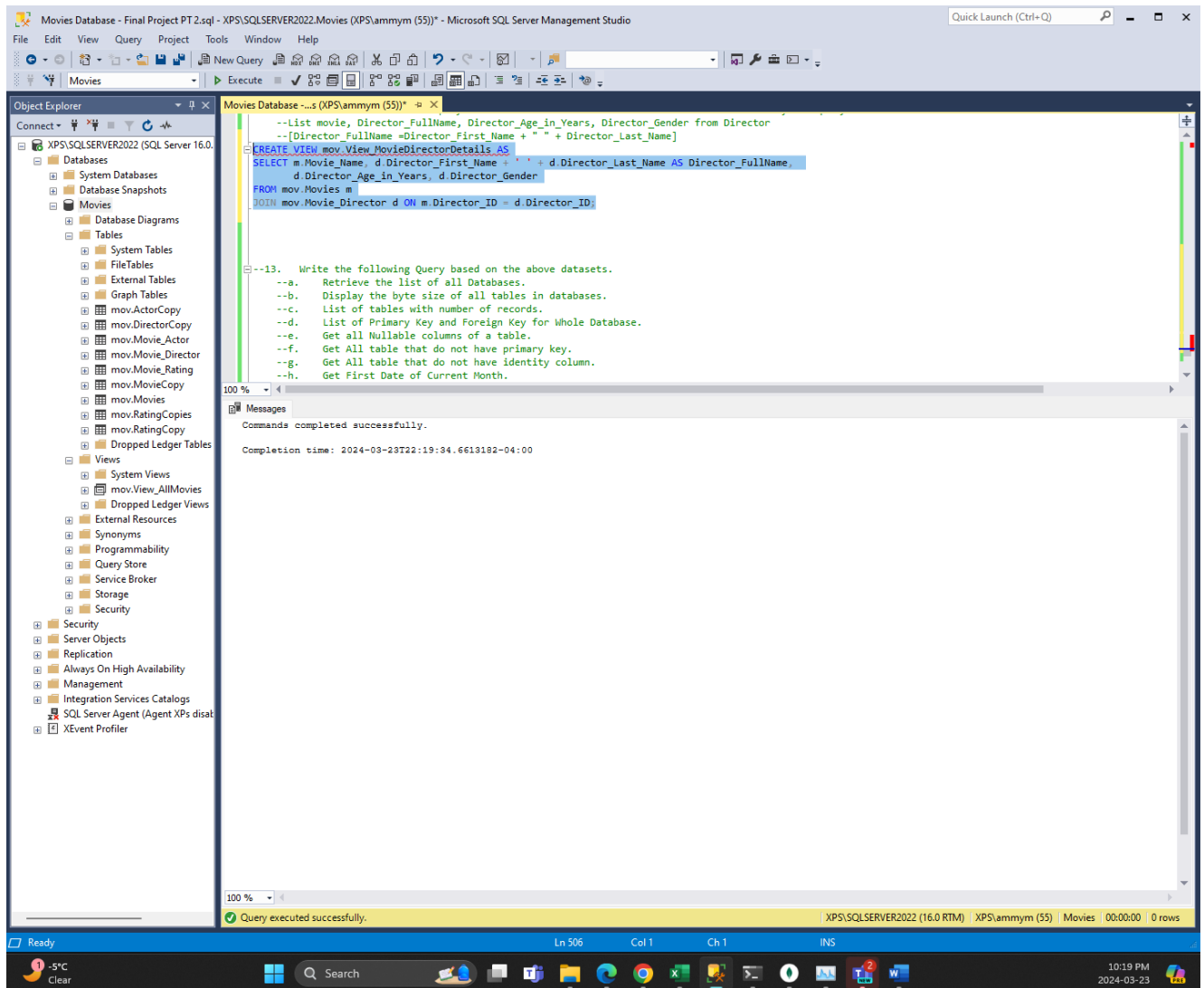
The right pane shows the 'Messages' window with the following text:

```
Commands completed successfully.
Completion time: 2024-03-23T22:13:17.1466643-04:00
```

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) XPS\ammym (55) | Movies | 00:00:00 | 0 rows'.

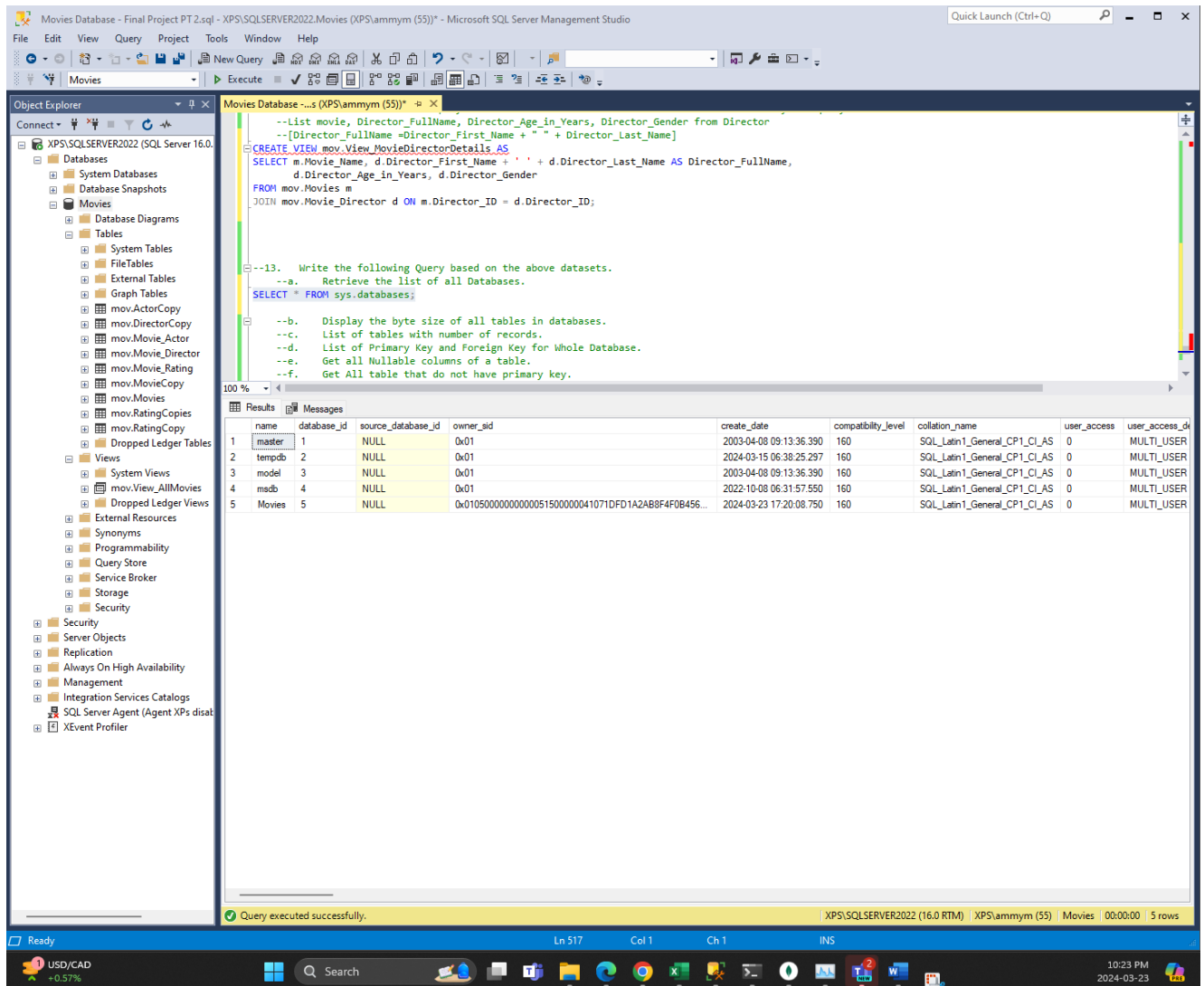
e. Create a view to display all the information based on the result set returned by the query as shown below-

List movie, Director_FullName, Director_Age_in_Years, Director_Gender from Director
[Director_FullName = Director_First_Name + " " + Director_Last_Name]



11. Write the following Query based on the above datasets.

a. Retrieve the list of all Databases.



b. Display the byte size of all tables in databases.

c. List of tables with number of records.

d. List of Primary Key and Foreign Key for Whole Database.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a SQL query window with the following code:

```
sys.allocation_units a ON p.partition_id = a.container_id
LEFT OUTER JOIN
sys.schemas s ON t.schema_id = s.schema_id
GROUP BY
t.Name, s.Name, p.Rows
ORDER BY
t.Name;

--c. List of tables with number of records.

--d. List of Primary Key and Foreign Key for Whole Database.
SELECT Constraint_Name AS [KeyList], *
FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE

--e. Get all Nullable columns of a table.
SELECT TABLE_NAME, * FROM INFORMATION_SCHEMA.COLUMNS
WHERE IS_NULLABLE = 'YES';
```

The bottom pane shows the 'Results' tab with a table containing 7 rows of data:

KeyList	CONSTRAINT_CATALOG	CONSTRAINT_SCHEMA	CONSTRAINT_NAME	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME	ORDINAL_POSITION
1 FK_Movie_ID	Movies	mov	FK_Movie_ID	Movies	mov	Movie_Actor	Movie_ID	1
2 FK_Movie_Rating_Movie_ID	Movies	mov	FK_Movie_Rating_Movie_ID	Movies	mov	Movie_Rating	Movie_ID	1
3 FK_Movies_Director_ID	Movies	mov	FK_Movies_Director_ID	Movies	mov	Movies	Director_ID	1
4 PK_Movie_Actor	Movies	mov	PK_Movie_Actor	Movies	mov	Movie_Actor	Actor_ID	1
5 PK_Movie_Director	Movies	mov	PK_Movie_Director	Movies	mov	Movie_Director	Director_ID	1
6 PK_Movie_Rating	Movies	mov	PK_Movie_Rating	Movies	mov	Movie_Rating	Movie_Rating_ID	1
7 PK_Movies	Movies	mov	PK_Movies	Movies	mov	Movies	Movie_ID	1

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) XPS\ammym (55) Movies 00:00:01 7 rows'.

e. Get all Nullable columns of a table.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The Object Explorer on the left shows the 'Movies' database selected. The central pane shows a SQL query executed successfully, which lists all nullable columns in the database. The query uses the INFORMATION_SCHEMA.COLUMNS table and filters for IS_NULLABLE = 'YES'. The results pane at the bottom shows a table with 10 columns: TABLE_NAME, TABLE_CATALOG, TABLE_SCHEMA, TABLE_NAME, COLUMN_NAME, ORDINAL_POSITION, COLUMN_DEFAULT, IS_NULLABLE, DATA_TYPE, and CHARACTER_MAXIMUM_LENGTH. The table contains 29 rows of data, listing various tables and their columns, including Movie_Director, Movie, Movie_Copy, and Movie_Rating.

```
sys.allocation_units a ON p.partition_id = a.container_id
LEFT OUTER JOIN
sys.schemas s ON t.schema_id = s.schema_id
GROUP BY
t.Name, s.Name, p.Rows
ORDER BY
t.Name;

--c. List of tables with number of records.
--d. List of Primary Key and Foreign Key for Whole Database.
--e. Get all Nullable columns of a table.

SELECT TABLE_NAME, * FROM INFORMATION_SCHEMA.COLUMNS
WHERE IS_NULLABLE = 'YES';

--f. Get all table that do not have primary key.
--SELECT * FROM sys.indexes
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.indexes i WHERE i.object_id = t.object_id AND i.is_primary_key = 1);
```

	TABLE_NAME	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME	ORDINAL_POSITION	COLUMN_DEFAULT	IS_NULLABLE	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH
1	Movie_Director	Movies	mov	Movie_Director	Director_First_Name	2	NULL	YES	varchar	50
2	Movie_Director	Movies	mov	Movie_Director	Director_Last_Name	3	NULL	YES	varchar	50
3	Movie_Director	Movies	mov	Movie_Director	Director_Age_in_Years	4	NULL	YES	int	NULL
4	Movie_Director	Movies	mov	Movie_Director	Director_Gender	5	NULL	YES	varchar	10
5	Movie	Movies	mov	Movie	Movie_Name	2	NULL	YES	varchar	50
6	Movie	Movies	mov	Movie	Movie_Released_Year	3	NULL	YES	int	NULL
7	Movie	Movies	mov	Movie	Movie_Lead_Studio	4	NULL	YES	varchar	50
8	Movie	Movies	mov	Movie	Movie_Language	5	NULL	YES	varchar	50
9	Movie	Movies	mov	Movie	Movie_Category	6	NULL	YES	varchar	50
10	Movie	Movies	mov	Movie	Movie_Duration_in_Min	7	NULL	YES	int	NULL
11	Movie	Movies	mov	Movie	Movie_Worldwide_Earning_in_\$M	8	NULL	YES	decimal	NULL
12	Movie	Movies	mov	Movie	Movie_Type	9	NULL	YES	varchar	50
13	Movie	Movies	mov	Movie	Director_ID	10	NULL	YES	int	NULL
14	Movie_Actor	Movies	mov	Movie_Actor	Actor_First_Name	2	NULL	YES	varchar	50
15	Movie_Actor	Movies	mov	Movie_Actor	Actor_Last_Name	3	NULL	YES	varchar	50
16	Movie_Actor	Movies	mov	Movie_Actor	Actor_Age_in_Years	4	NULL	YES	int	NULL
17	Movie_Actor	Movies	mov	Movie_Actor	Actor_Location	5	NULL	YES	varchar	50
18	Movie_Actor	Movies	mov	Movie_Actor	Movie_ID	6	NULL	YES	int	NULL
19	Movie_Rating	Movies	mov	Movie_Rating	Rating_Audience_Score	2	NULL	YES	varchar	10
20	Movie_Rating	Movies	mov	Movie_Rating	Rating_Rotten_Tomatoes	3	NULL	YES	varchar	10
21	Movie_Rating	Movies	mov	Movie_Rating	Movie_ID	4	NULL	YES	int	NULL
22	Movie_Copy	Movies	mov	Movie_Copy	Movie_Name	2	NULL	YES	varchar	50
23	Movie_Copy	Movies	mov	Movie_Copy	Movie_Released_Year	3	NULL	YES	int	NULL
24	Movie_Copy	Movies	mov	Movie_Copy	Movie_Lead_Studio	4	NULL	YES	varchar	50
25	Movie_Copy	Movies	mov	Movie_Copy	Movie_Language	5	NULL	YES	varchar	50
26	Movie_Copy	Movies	mov	Movie_Copy	Movie_Category	6	NULL	YES	varchar	50
27	Movie_Copy	Movies	mov	Movie_Copy	Movie_Duration_in_Min	7	NULL	YES	int	NULL
28	Movie_Copy	Movies	mov	Movie_Copy	Movie_Worldwide_Earning_in_\$M	8	NULL	YES	decimal	NULL
29	Movie_Copy	Movies	mov	Movie_Copy	Movie_Type	9	NULL	YES	varchar	50

Microsoft SQL Server Management Studio interface showing a query execution for the 'Movies' database.

Object Explorer: Shows the database structure including Databases, Tables, Views, and Security.

Query Window: Contains the following SQL query:

```
--c. List of tables with number of records.
--d. List of Primary Key and Foreign Key for Whole Database.
--e. Get all Nullable columns of a table.

SELECT * FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Movies' AND IS_NULLABLE = 'YES';

--f. Get All table that do not have primary key.
--g. Get All table that do not have identity column.
--h. Get First Date of Current Month.
--i. Get Last date of Current month.
--j. Get first date of next month.
--k. Get Last date of next month.
--l. Get all the information from the tables.
--m. Get all columns contain any constraints.
--n. Get all tables that contain a view.
--o. Get all columns of table that using in views.
```

Results: The query returned 9 rows of data from the INFORMATION_SCHEMA.COLUMNS table.

	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME	ORDINAL_POSITION	COLUMN_DEFAULT	IS_NULLABLE	DATA_TYPE	CHARACTER_MAXIMUM_LENGTH	CHARACTER_OCTET_LENGTH
1	mov	mov	Movies	Movie_Name	2	NULL	YES	varchar	50	50
2	mov	mov	Movies	Movie_Released_Year	3	NULL	YES	int	NULL	NULL
3	mov	mov	Movies	Movie_Lead_Studio	4	NULL	YES	varchar	50	50
4	mov	mov	Movies	Movie_Language	5	NULL	YES	varchar	50	50
5	mov	mov	Movies	Movie_Category	6	NULL	YES	varchar	50	50
6	mov	mov	Movies	Movie_Duration_In_Min	7	NULL	YES	int	NULL	NULL
7	mov	mov	Movies	Movie_Worldwide_Earning_In_M	8	NULL	YES	decimal	NULL	NULL
8	mov	mov	Movies	Movie_Type	9	NULL	YES	varchar	50	50
9	mov	mov	Movies	Director_ID	10	NULL	YES	int	NULL	NULL

Query executed successfully.

f. Get All table that do not have primary key.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'Movies Database - Final Project PT 2.sql'. The main query window contains the following SQL code:

```
--c. List of tables with number of records.
--d. List of Primary Key and Foreign Key for Whole Database.
--e. Get all Nullable columns of a table.

SELECT * FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Movies' AND IS_NULLABLE = 'YES';

--f. Get All table that do not have primary key.
--SELECT * FROM sys.indexes
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.indexes i WHERE i.object_id = t.object_id AND i.is_primary_key = 1);

--g. Get All table that do not have identity column.
--h. Get First Date of Current Month.
--i. Get Last date of Current month.
--j. Get first date of next month.
--k. Get Last date of next month.
--l. Get all the information from the tables.
```

The Results pane shows the output of the query, listing tables without primary keys:

	name	object_id	principal_id	schema_id	parent_object_id	type	type_desc	create_date	modify_date	is_ms_shipped	is_published	is_schema_published	lob_data_space
1	MovieCopy	1173579219	NULL	5	0	U	USER_TABLE	2024-03-23 21:29:05.833	2024-03-23 21:29:05.833	0	0	0	0
2	DirectorCopy	1189579276	NULL	5	0	U	USER_TABLE	2024-03-23 21:30:28.623	2024-03-23 21:30:28.623	0	0	0	0
3	ActorCopy	1205579333	NULL	5	0	U	USER_TABLE	2024-03-23 21:31:58.077	2024-03-23 21:31:58.077	0	0	0	0
4	RatingCopy	1221579390	NULL	5	0	U	USER_TABLE	2024-03-23 21:33:09.950	2024-03-23 21:33:09.950	0	0	0	0
5	RatingCopies	1237579447	NULL	5	0	U	USER_TABLE	2024-03-23 21:34:04.790	2024-03-23 21:34:04.790	0	0	0	0

The status bar at the bottom indicates 'Query executed successfully.' and shows the current file path: 'XPS\SQLSERVER2022 (16.0 RTM) XPS\ammym (55) Movies 00:00:00 5 rows'.

g. Get All table that do not have identity column.

The screenshot shows the Microsoft SQL Server Management Studio (SSMS) interface. The left pane displays the Object Explorer with the 'Movies' database selected. The central pane shows a SQL query window with the following code:

```
--SELECT * FROM sys.indexes
--SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.indexes i WHERE i.object_id = t.object_id AND i.is_primary_key = 1);

--g. Get All table that do not have identity column.
--SELECT * FROM sys.columns
--Where is_identity =1;
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.columns c WHERE c.object_id = t.object_id AND c.is_identity = 1);

--h. Get First Date of Current Month.
--i. Get Last date of Current month.
--j. Get first date of next month.
--k. Get Last date of next month.
--l. Get all the information from the tables.
--m. Get all columns contain any constraints.
--n. Get all tables that contain a view.
```

The bottom pane shows the Results tab with a table structure. The status bar at the bottom indicates 'Query executed successfully.' and '0 rows'.

name	object_id	principal_id	schema_id	parent_object_id	type	type_desc	create_date	modify_date	is_ms_shipped	is_published	is_schema_published	lob_data_space_id	filestream_data_space_id	max_c
------	-----------	--------------	-----------	------------------	------	-----------	-------------	-------------	---------------	--------------	---------------------	-------------------	--------------------------	-------

h. Get First Date of Current Month.

The screenshot displays the Microsoft SQL Server Management Studio interface. The Object Explorer on the left shows the database structure for 'XPS\SQLSERVER2022 (SQL Server 16.0)'. The main query window contains the following SQL code:

```
--SELECT * FROM sys.indexes
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.indexes i WHERE i.object_id = t.object_id AND i.is_primary_key = 1);

--g. Get All table that do not have identity column.
--SELECT * FROM sys.columns
--Where is_identity =1;
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.columns c WHERE c.object_id = t.object_id AND c.is_identity = 1);

--h. Get First Date of Current Month.
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()), 0) AS FirstDayOfMonth;

--i. Get Last date of Current month.
--j. Get first date of next month.
--k. Get Last date of next month.
--l. Get all the information from the tables.
```

The Results pane at the bottom shows the output of the query:

FirstDayOfMonth
2024-03-01 00:00:00.000

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 1 rows'.

i. Get Last date of Current month.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The right pane shows a query window with the following SQL code:

```
--SELECT * FROM sys.columns
--where is_identity =1;
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.columns c WHERE c.object_id = t.object_id AND c.is_identity = 1);

--h. Get First Date of Current Month.
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()), 0) AS FirstDayOfMonth;

--i. Get Last date of Current month.
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;

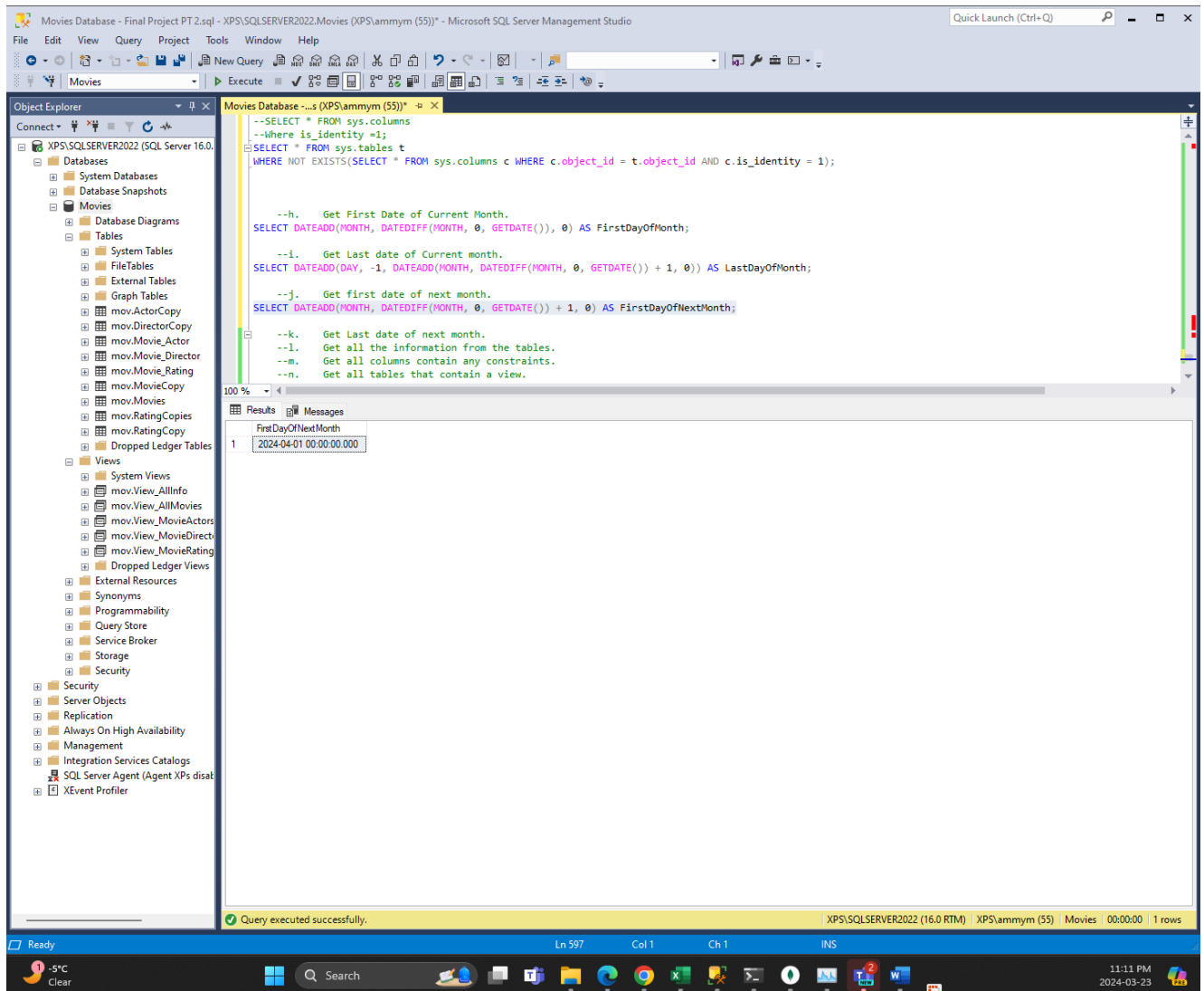
--j. Get first date of next month.
--k. Get Last date of next month.
--l. Get all the information from the tables.
--m. Get all columns contain any constraints.
--n. Get all tables that contain a view.
--o. Get all columns of table that using in views.
```

The 'Results' pane shows the output of the query:

1	LastDayOfMonth
1	2024-03-31 00:00:00.000

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 1 rows'.

j. Get first date of next month.



The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a query window with the following SQL code:

```
--SELECT * FROM sys.columns
--where is_identity =1;
SELECT * FROM sys.tables t
WHERE NOT EXISTS(SELECT * FROM sys.columns c WHERE c.object_id = t.object_id AND c.is_identity = 1);

--h. Get First Date of Current Month.
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()), 0) AS FirstDayOfMonth;

--i. Get Last date of Current month.
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;

--j. Get first date of next month.
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0) AS FirstDayOfNextMonth;

--k. Get Last date of next month.
--l. Get all the information from the tables.
--m. Get all columns contain any constraints.
--n. Get all tables that contain a view.
```

The right pane shows the 'Results' tab with a single row of data:

FirstDayOfNextMonth
2024-04-01 00:00:00.000

The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 1 rows'.

k. Get Last date of next month.

The screenshot displays the Microsoft SQL Server Management Studio (SSMS) interface. The title bar indicates the connection is to 'Movies Database - Final Project PT 2.sql - XPS\SQLSERVER2022\Movies (XPS\ammym (55))'. The left-hand 'Object Explorer' pane shows the database structure, including 'Databases', 'System Databases', 'Database Snapshots', 'Movies', 'Database Diagrams', 'Tables', 'System Tables', 'FileTables', 'External Tables', 'Graph Tables', 'mov.ActorCopy', 'mov.DirectorCopy', 'mov.Movie_Actor', 'mov.Movie_Director', 'mov.Movie_Rating', 'mov.MovieCopy', 'mov.Movies', 'mov.RatingCopies', 'mov.RatingCopy', 'Dropped Ledger Tables', 'Views', 'System Views', 'mov.View_AllInfo', 'mov.View_AllMovies', 'mov.View_MovieActors', 'mov.View_MovieDirect', 'mov.View_MovieRating', 'Dropped Ledger Views', 'External Resources', 'Synonyms', 'Programmability', 'Query Store', 'Service Broker', 'Storage', 'Security', 'Server Objects', 'Replication', 'Always On High Availability', 'Management', 'Integration Services Catalogs', 'SQL Server Agent (Agent XPs disabled)', and 'XEvent Profiler'.

The central 'Query Editor' window contains the following SQL script:

```
--h. Get First Date of Current Month.  
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()), 0) AS FirstDayOfMonth;  
  
--i. Get Last date of Current month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;  
  
--j. Get first date of next month.  
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0) AS FirstDayOfNextMonth;  
  
--k. Get Last date of next month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 2, 0)) AS LastDayofNextMonth;  
  
--l. Get all the information from the tables.  
--m. Get all columns contain any constraints.  
--n. Get all tables that contain a view.  
--o. Get all columns of table that using in views.
```

The 'Results' pane at the bottom shows the output of the query, displaying a single row with the value '2024-04-30 00:00:00.000' under the column header 'LastDayofNextMonth'.

The status bar at the bottom indicates 'Query executed successfully.' and provides details about the connection: 'XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 1 rows'.

I. Get all the information from the tables.

The screenshot displays the Microsoft SQL Server Management Studio interface. The left pane shows the Object Explorer with the 'Movies' database selected. The central query editor contains the following SQL script:

```
--i. Get Last date of Current month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;  
  
--j. Get first date of next month.  
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0) AS FirstDayOfNextMonth;  
  
--k. Get Last date of next month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 2, 0)) AS LastDayOfNextMonth;  
  
--l. Get all the information from the tables.  
SELECT * FROM sys.tables  
  
--m. Get all columns contain any constraints.  
SELECT * FROM INFORMATION_SCHEMA.CONSTRAINT_COLUMN_USAGE;  
  
--n. Get all tables that contain a view.  
--o. Get all columns of table that using in views.
```

The bottom pane shows the 'Results' tab with a table containing 9 rows of data. The status bar at the bottom indicates 'Query executed successfully.' and '9 rows'.

	name	object_id	principal_id	schema_id	parent_object_id	type	type_desc	create_date	modify_date	is_ms_shipped	is_published	is_schema_published	lob_data_sp
1	Movie_Director	901578250	NULL	5	0	U	USER_TABLE	2024-03-23 19:24:05.700	2024-03-23 19:51:40.190	0	0	0	0
2	Movies	949578421	NULL	5	0	U	USER_TABLE	2024-03-23 19:51:40.173	2024-03-23 20:35:26.873	0	0	0	0
3	Movie_Actor	1013578649	NULL	5	0	U	USER_TABLE	2024-03-23 20:22:26.473	2024-03-23 20:22:26.473	0	0	0	0
4	Movie_Rating	1109578991	NULL	5	0	U	USER_TABLE	2024-03-23 20:35:26.860	2024-03-23 20:35:26.860	0	0	0	0
5	MovieCopy	1173579219	NULL	5	0	U	USER_TABLE	2024-03-23 21:29:05.833	2024-03-23 21:29:05.833	0	0	0	0
6	DirectorCopy	1189579276	NULL	5	0	U	USER_TABLE	2024-03-23 21:30:28.623	2024-03-23 21:30:28.623	0	0	0	0
7	ActorCopy	1205579333	NULL	5	0	U	USER_TABLE	2024-03-23 21:31:58.077	2024-03-23 21:31:58.077	0	0	0	0
8	RatingCopy	1221579390	NULL	5	0	U	USER_TABLE	2024-03-23 21:33:09.950	2024-03-23 21:33:09.950	0	0	0	0
9	RatingCopies	1237579447	NULL	5	0	U	USER_TABLE	2024-03-23 21:34:04.790	2024-03-23 21:34:04.790	0	0	0	0

m. Get all columns contain any constraints.

The screenshot displays the Microsoft SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'Movies' database selected. The central pane shows a SQL query window with the following text:

```
--i. Get Last date of Current month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;  
  
--j. Get first date of next month.  
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0) AS FirstDayOfNextMonth;  
  
--k. Get Last date of next month.  
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 2, 0)) AS LastDayofNextMonth;  
  
--l. Get all the information from the tables.  
SELECT * FROM sys.tables  
  
--m. Get all columns contain any constraints.  
SELECT * FROM INFORMATION_SCHEMA.CONSTRAINT_COLUMN_USAGE;  
  
--n. Get all tables that contain a view.  
--o. Get all columns of table that using in views.
```

The bottom pane shows the 'Results' tab with a table of 7 rows and 7 columns. The status bar at the bottom indicates 'Query executed successfully.' and 'XPS\SQLSERVER2022 (16.0 RTM) XPS\ammym (55) Movies 00:00:00 7 rows'.

	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME	CONSTRAINT_CATALOG	CONSTRAINT_SCHEMA	CONSTRAINT_NAME
1	Movies	mov	Movie_Actor	Movie_ID	Movies	mov	FK_Movie_ID
2	Movies	mov	Movie_Rating	Movie_ID	Movies	mov	FK_Movie_Rating_Movie_ID
3	Movies	mov	Movies	Director_ID	Movies	mov	FK_Movies_Director_ID
4	Movies	mov	Movie_Actor	Actor_ID	Movies	mov	PK_Movie_Actor
5	Movies	mov	Movie_Director	Director_ID	Movies	mov	PK_Movie_Director
6	Movies	mov	Movie_Rating	Movie_Rating_ID	Movies	mov	PK_Movie_Rating
7	Movies	mov	Movies	Movie_ID	Movies	mov	PK_Movies

n. Get all tables that contain a view.

The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the 'Movies' database structure, including tables, views, and other database objects. The central query window contains a SQL script with several queries, including one to find tables containing views. The 'Results' pane at the bottom displays the output of the query, showing a list of tables and their associated views.

```
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0)) AS LastDayOfMonth;

--j. Get first date of next month.
SELECT DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 1, 0) AS FirstDayOfNextMonth;

--k. Get Last date of next month.
SELECT DATEADD(DAY, -1, DATEADD(MONTH, DATEDIFF(MONTH, 0, GETDATE()) + 2, 0)) AS LastDayOfNextMonth;

--l. Get all the information from the tables.
SELECT * FROM sys.tables;

--m. Get all columns contain any constraints.
SELECT * FROM INFORMATION_SCHEMA.CONSTRAINT_COLUMN_USAGE;

--n. Get all tables that contain a view.
SELECT * FROM INFORMATION_SCHEMA.VIEW_TABLE_USAGE;

--o. Get all columns of table that using in views.
```

	VIEW_CATALOG	VIEW_SCHEMA	VIEW_NAME	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME
1	Movies	mov	View_AllInfo	Movies	mov	Movie_Actor
2	Movies	mov	View_AllInfo	Movies	mov	Movie_Director
3	Movies	mov	View_AllInfo	Movies	mov	Movie_Rating
4	Movies	mov	View_AllInfo	Movies	mov	Movies
5	Movies	mov	View_AllMovies	Movies	mov	Movies
6	Movies	mov	View_MovieActors	Movies	mov	Movie_Actor
7	Movies	mov	View_MovieActors	Movies	mov	Movies
8	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director
9	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movies
10	Movies	mov	View_MovieRatings	Movies	mov	Movie_Rating
11	Movies	mov	View_MovieRatings	Movies	mov	Movies

Query executed successfully. XPS\SQLSERVER2022 (16.0 RTM) | XPS\ammym (55) | Movies | 00:00:00 | 11 rows

o. Get all columns of table that using in views.

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with the 'Movies' database selected. The right pane shows a query window with the following SQL code:

```
--1. Get all the information from the tables.
SELECT * FROM sys.tables;

--m. Get all columns contain any constraints.
SELECT * FROM INFORMATION_SCHEMA.CONSTRAINT_COLUMN_USAGE;

--n. Get all tables that contain a view.
SELECT * FROM INFORMATION_SCHEMA.VIEW_TABLE_USAGE;

--o. Get all columns of table that using in views.
SELECT COLUMN_NAME, TABLE_NAME, *
FROM INFORMATION_SCHEMA.VIEW_COLUMN_USAGE;
```

The 'Results' pane displays the output of the query, showing a list of columns and tables used in views. The status bar at the bottom indicates 'Query executed successfully.' and '43 rows'.

1	COLUMN_NAME	TABLE_NAME	VIEW_CATALOG	VIEW_SCHEMA	VIEW_NAME	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	COLUMN_NAME
1	Director_ID	Movie_Director	Movies	mov	View_AllInfo	Movies	mov	Movie_Director	Director_ID
2	Director_ID	Movie_Director	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director	Director_ID
3	Director_First_Name	Movie_Director	Movies	mov	View_AllInfo	Movies	mov	Movie_Director	Director_First_Name
4	Director_First_Name	Movie_Director	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director	Director_First_Name
5	Director_Last_Name	Movie_Director	Movies	mov	View_AllInfo	Movies	mov	Movie_Director	Director_Last_Name
6	Director_Last_Name	Movie_Director	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director	Director_Last_Name
7	Director_Age_in_Years	Movie_Director	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director	Director_Age_in_Years
8	Director_Gender	Movie_Director	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movie_Director	Director_Gender
9	Movie_ID	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_ID
10	Movie_ID	Movies	Movies	mov	View_MovieRatings	Movies	mov	Movies	Movie_ID
11	Movie_ID	Movies	Movies	mov	View_MovieActors	Movies	mov	Movies	Movie_ID
12	Movie_ID	Movies	Movies	mov	View_AllInfo	Movies	mov	Movies	Movie_ID
13	Movie_Name	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Name
14	Movie_Name	Movies	Movies	mov	View_MovieRatings	Movies	mov	Movies	Movie_Name
15	Movie_Name	Movies	Movies	mov	View_MovieActors	Movies	mov	Movies	Movie_Name
16	Movie_Name	Movies	Movies	mov	View_AllInfo	Movies	mov	Movies	Movie_Name
17	Movie_Name	Movies	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movies	Movie_Name
18	Movie_Released_Year	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Released_Year
19	Movie_Released_Year	Movies	Movies	mov	View_MovieRatings	Movies	mov	Movies	Movie_Released_Year
20	Movie_Lead_Studio	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Lead_Studio
21	Movie_Language	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Language
22	Movie_Category	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Category
23	Movie_Duration_in_Min	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Duration_in_Min
24	Movie_Worldwide_Earning_in_\$M	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Worldwide_Earning_in_\$M
25	Movie_Type	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Movie_Type
26	Director_ID	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	Director_ID
27	Director_ID	Movies	Movies	mov	View_AllInfo	Movies	mov	Movies	Director_ID
28	Director_ID	Movies	Movies	mov	View_MovieDirectorDetails	Movies	mov	Movies	Director_ID
29	CreatedOn	Movies	Movies	mov	View_AllMovies	Movies	mov	Movies	CreatedOn
30	Actor_First_Name	Movie_Actor	Movies	mov	View_MovieActors	Movies	mov	Movie_Actor	Actor_First_Name